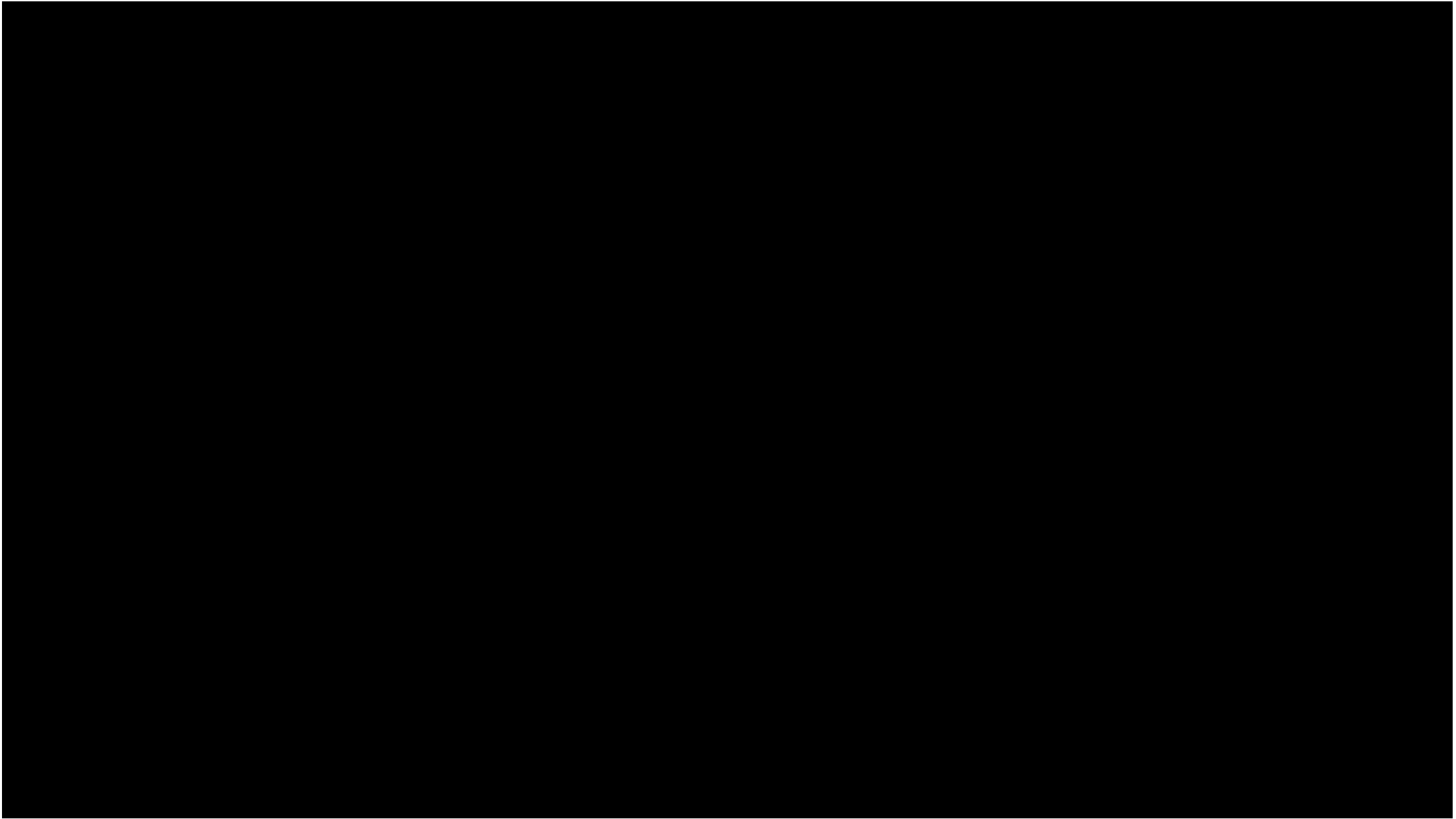




期末報告評分

- 期末競賽評分：所有同學評分 + TA與客座專家評分
- 當週作業成績：
 - 入選組別的口頭報告
 - 未入選組別的評論報告
- 入選組別的口頭報告獎勵：
 - 一等：學期等第 A +
 - 二等：學期等第 (至少) A
 - 三等：期末考+50分
 - 其他入選組別：期末考+10分
- 未入選組別的書面報告獎勵：
 - 一等：學期等第 A +
 - 二等：學期等第 (至少) A
 - 三等：期末考+50分

Grade

說明：

- 一、依據學位授予法第17條第1項規定，學校授予之學位，有入學資格或修業情形有不實或舞弊情事或論文、作品、成就證明、書面報告、技術報告或專業實務報告有造假、變造、抄襲、由他人代寫或其他舞弊情事之一者，應予撤銷，並公告註銷其已頒給之學位證書；另依專科以上學校學術倫理案件處理原則第6條第1項規定，學校應訂定學術倫理相關規定，包括學術倫理規範、權責單位、修習辦法、違反態樣、處理程序、處分條款及監管機制等規定，並依學校章則訂定程序辦理後，公告周知。為確保學校於接獲涉及學位授予法第17條情事檢舉案件，確實依學校處理程序進行審議，爰嗣後經本部函轉學校涉及學位授予法第17條情事之案件，請於案件審議完後填寫「專科以上學校學生學位論文涉及學位授予法第17條情事查核表

期中考

- 原始分數與等第轉換

- 85+ A+
- 70-85 A
- 65-69 A-
- 60-64 B+
- 55-59 B
- 50-54 B-
- 45-49 C+
- 40-44 C
- 35-39 C-

期末考

- 期末考範圍為整學期的內容
- 期中考未達C-者，若期末考分數及格或高於平均，則期中考等第改為C-

Beyond GI System: On-going GI Science

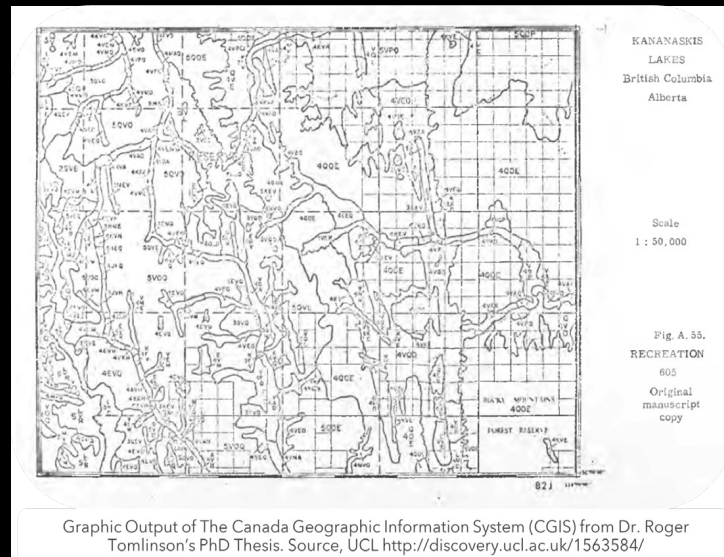
GEOG 1026

Department of Geography
National Taiwan University



The Emergence of GIS

- GIS = GI system = Geographic Information System
- The definition of GIS
 - GIS is fundamentally about the use of digital data to represent space and time (Longley et al., 2005).
- Roger Tomlinson (Father of GIS) developed first computerized GIS, Canadian Geographic Information System (CGIS) in 1963.



Graphic Output of The Canada Geographic Information System (CGIS) from Dr. Roger Tomlinson's PhD Thesis. Source, UCL <http://discovery.ucl.ac.uk/1563584/>

Fundamental Geo-Processing and GIS Analysis

- Spatial and attribute query
- Data transformation
- Data management
- Spatial extraction
- Spatial overlay
- Proximity analysis
- Spatial statistics analysis
- Data classification
- Spatial interpolation
- Terrain analysis
- Network analysis

GIS is not just software!

What geographic knowledge can we gain?

Evolution of GIS

GI System

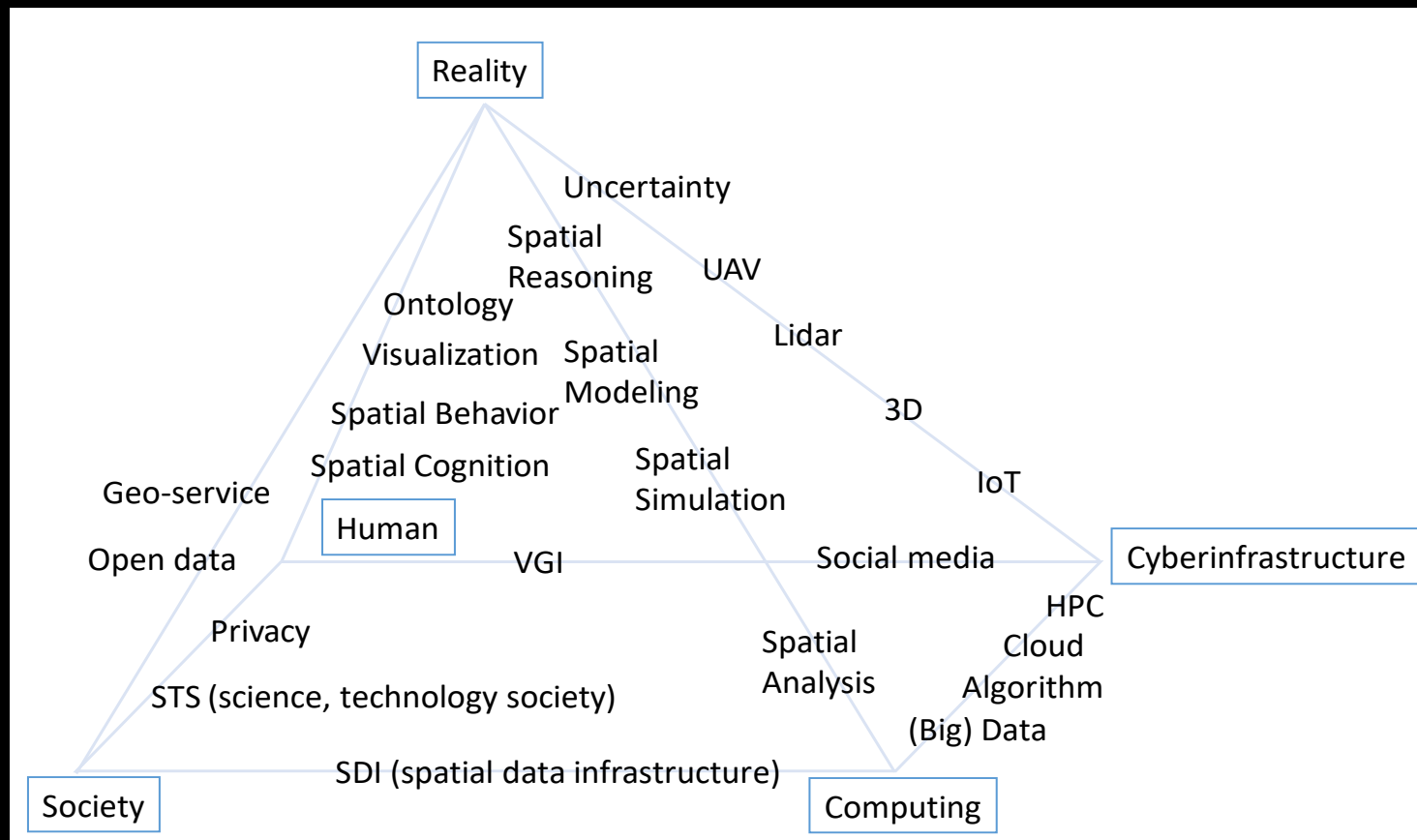
- Tool-oriented
- How do I use GIS software?
- How do I analyze GIS data?
- How do I plot a map?

GI Science

- Theory-oriented
- How to transfer reality into digital data?
- What geographic information can we know?
- How to facilitate geographic analysis?

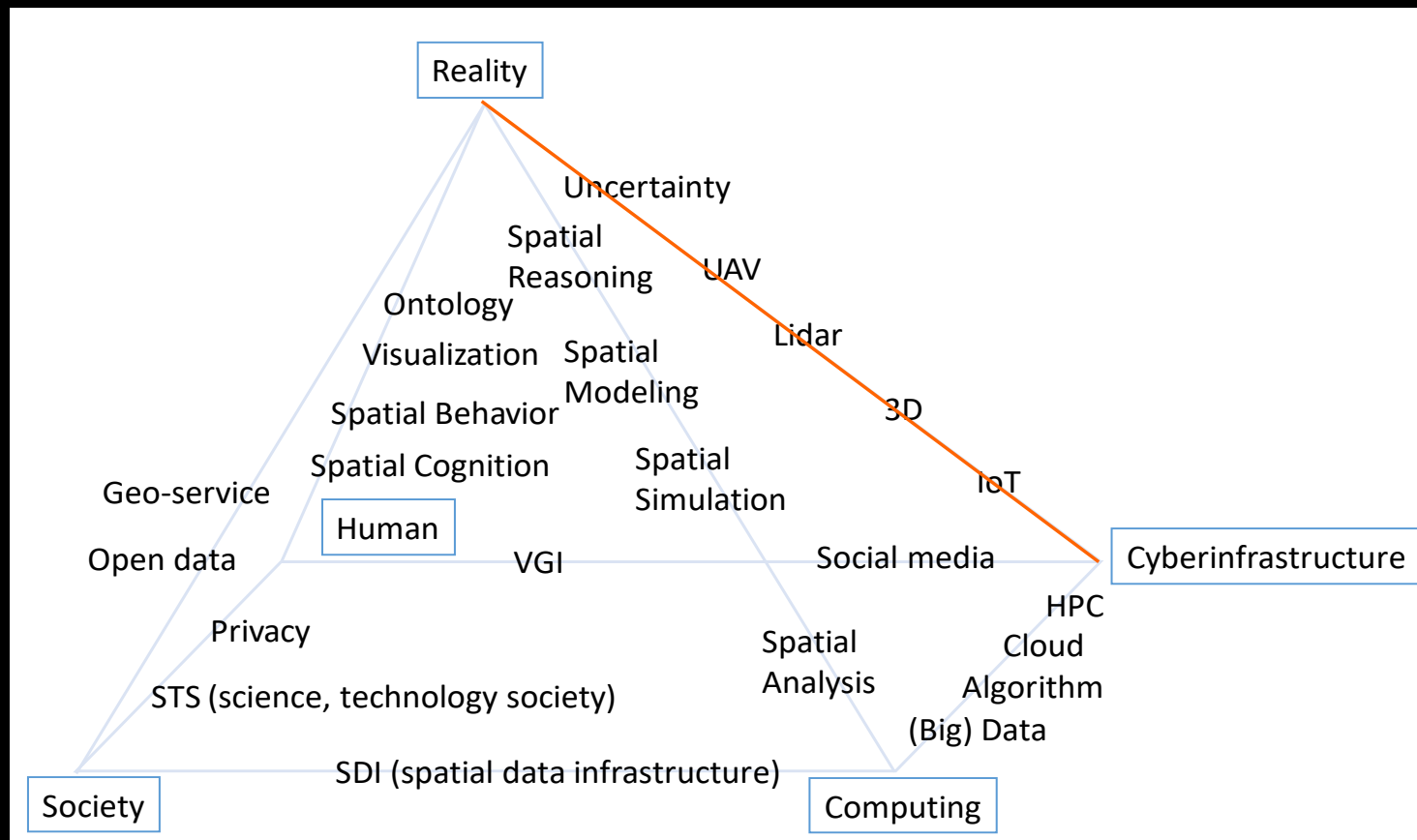
New Era of GIS

- GIS = GI Science = Geographic Information Science



New Era of GIS

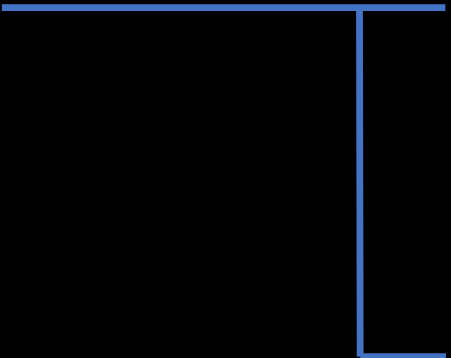
- GIS = GI Science = Geographic Information Science



Surveying and Mapping (測繪)

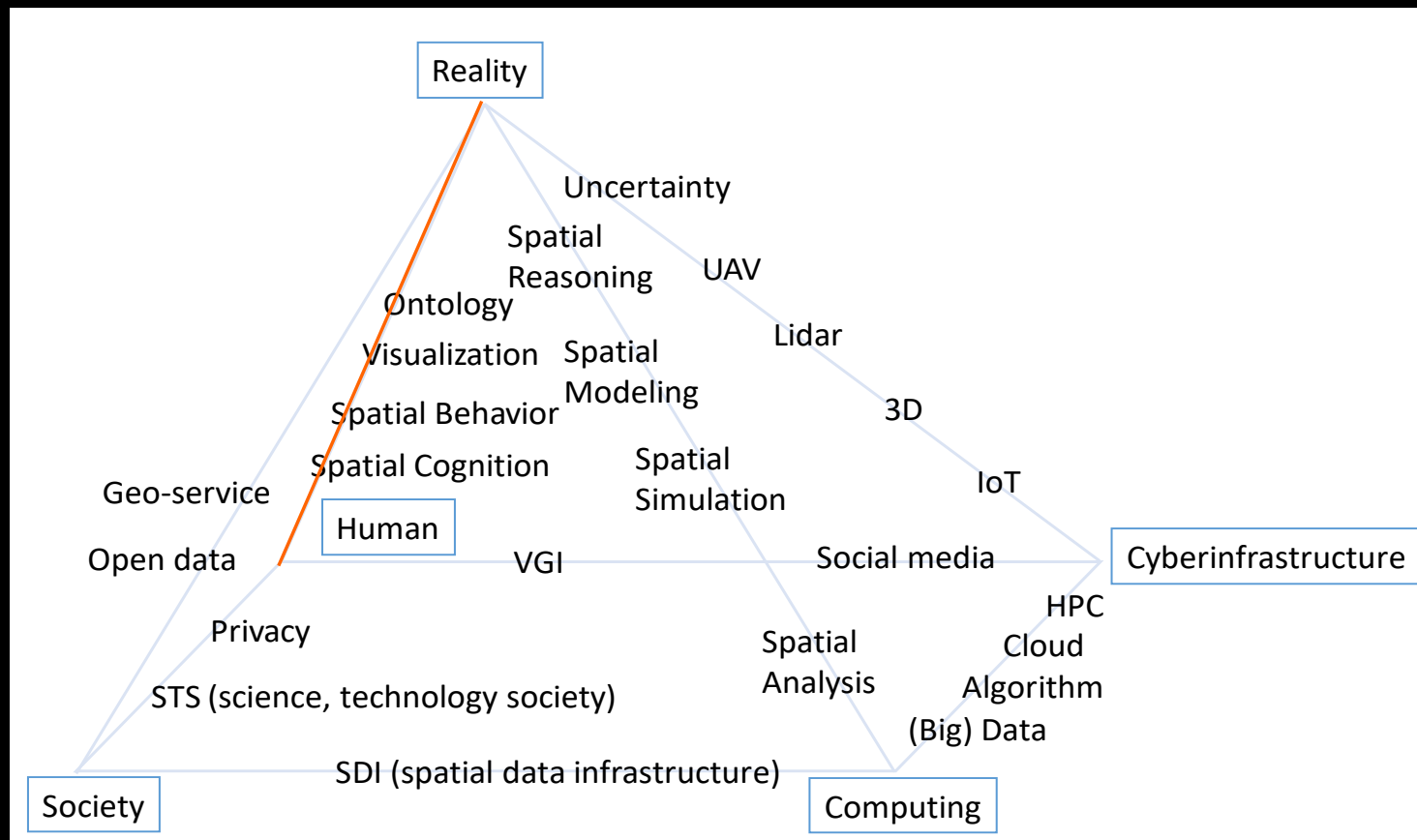
- Geodetic surveys
- Digitization from remote sensing imagery
- 3D models from LiDAR and photogrammetry (攝影測量)
- IoT sensors
- Uncertainty, error, and data quality

Specialty of Geospatial Data

- Data types
 - File format
 - Coordinates
 - Ontology
 - Uncertainty
 - Collection methods
- 
- Scale/resolution
 - Positional accuracy
 - Precision
 - Age of data
 - Metadata

New Era of GIS

- GIS = GI Science = Geographic Information Science

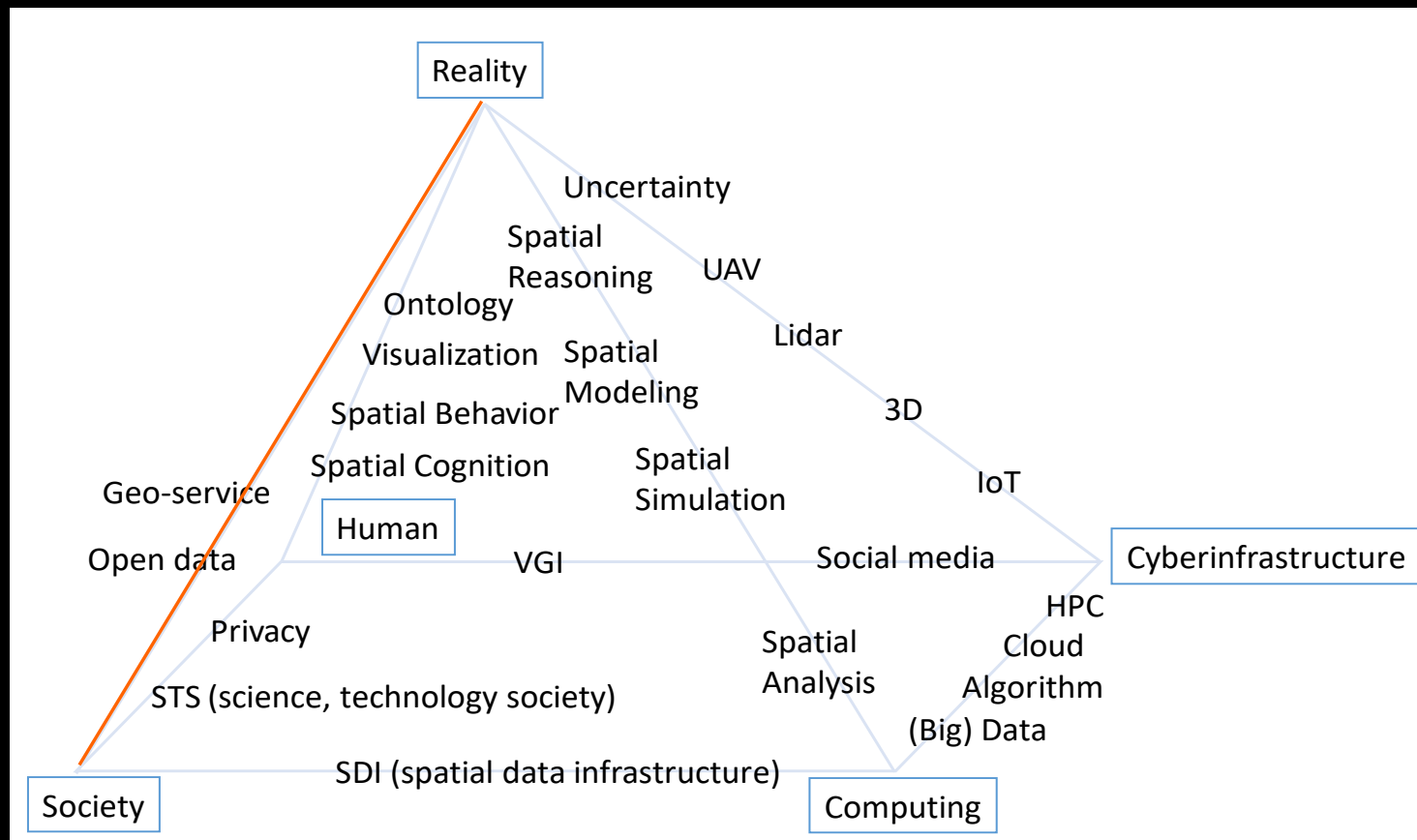


Ontology and Human Cognition

- How do we translate the real world into data?
- How humans perceive maps ➡ cartographic design principles
- Cognitive load in visualization ➡ visual variables and perception

New Era of GIS

- GIS = GI Science = Geographic Information Science



Location-Based Service (LBS)

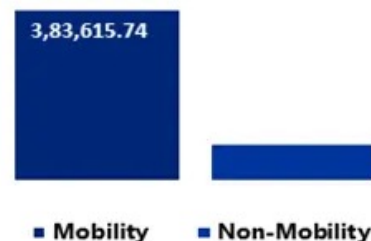
LBS (Location Based Services) Market Overview



Global Market Analysis, By Region, 2035

North America is anticipated to dominate the market from 2026 to 2035, owing to increasing smartphone adoption, growing demand for location-based applications, and a rising need for real-time location data across various industries.

Global Market Analysis, By Type (in USD million) 2035



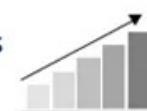
Key Players in the Market

- ALE International
- Apple INC
- Cisco Systems, Inc.
- Ericsson group
- Esri
- GL Communications Inc
- Glympse Inc.
- HERE
- Hewlett Packard Enterprise Development LP
- HID Global Corporation
- IBM
- IndoorAtlas

Growth Rate

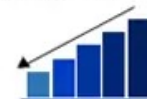


Growth Drivers



- Emerging Technologies for Location-Based Services
- Rise in 5G Network Rollout
- Growth of E-commerce and On-demand Services

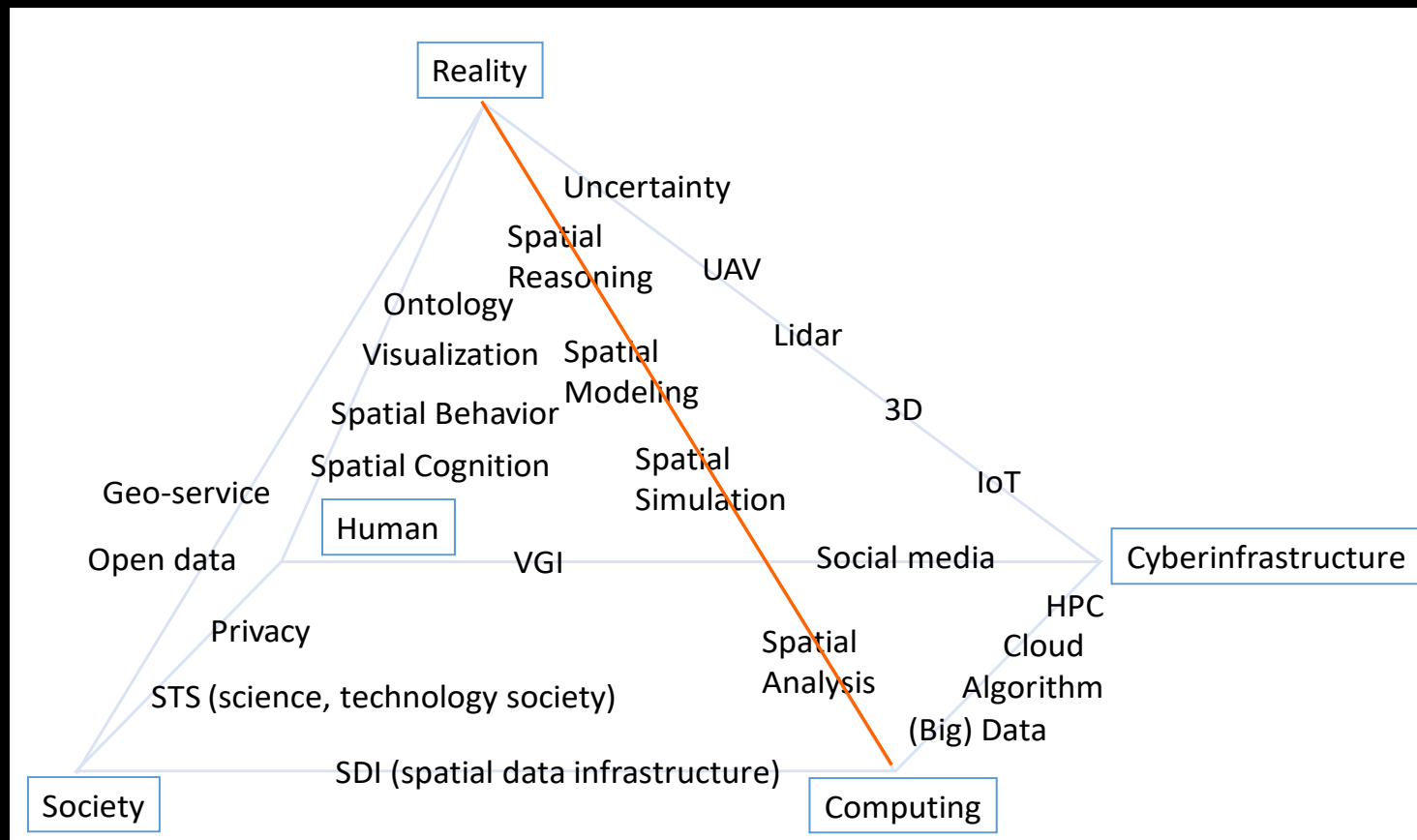
Challenges



- Privacy and Data Security Concerns
- High Cost of Infrastructure and Implementation

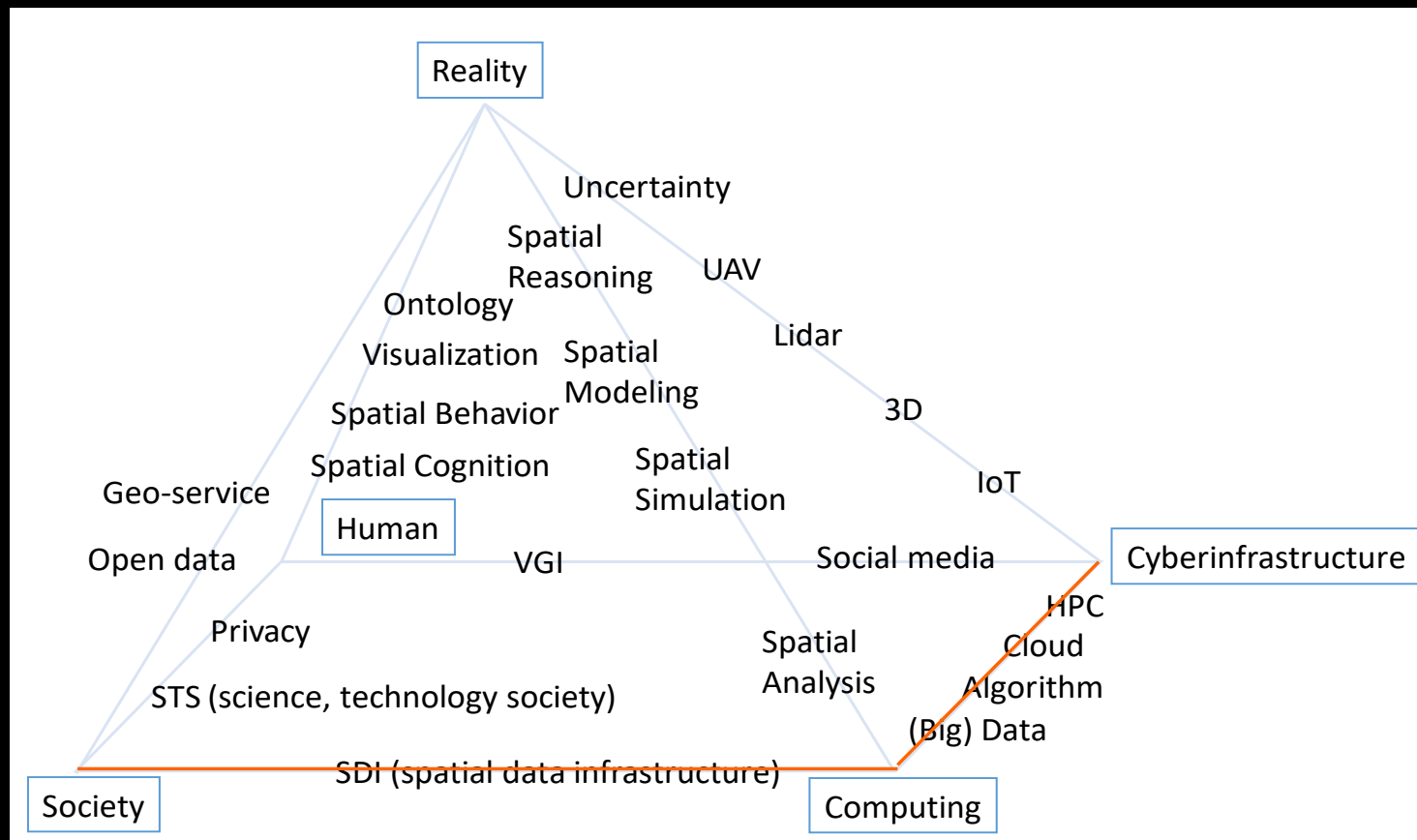
New Era of GIS

- GIS = GI Science = Geographic Information Science



New Era of GIS

- GIS = GI Science = Geographic Information Science



Geospatial Data Science and Engineering

**Geospatial Data
Engineering**

**Geospatial Data
Science**

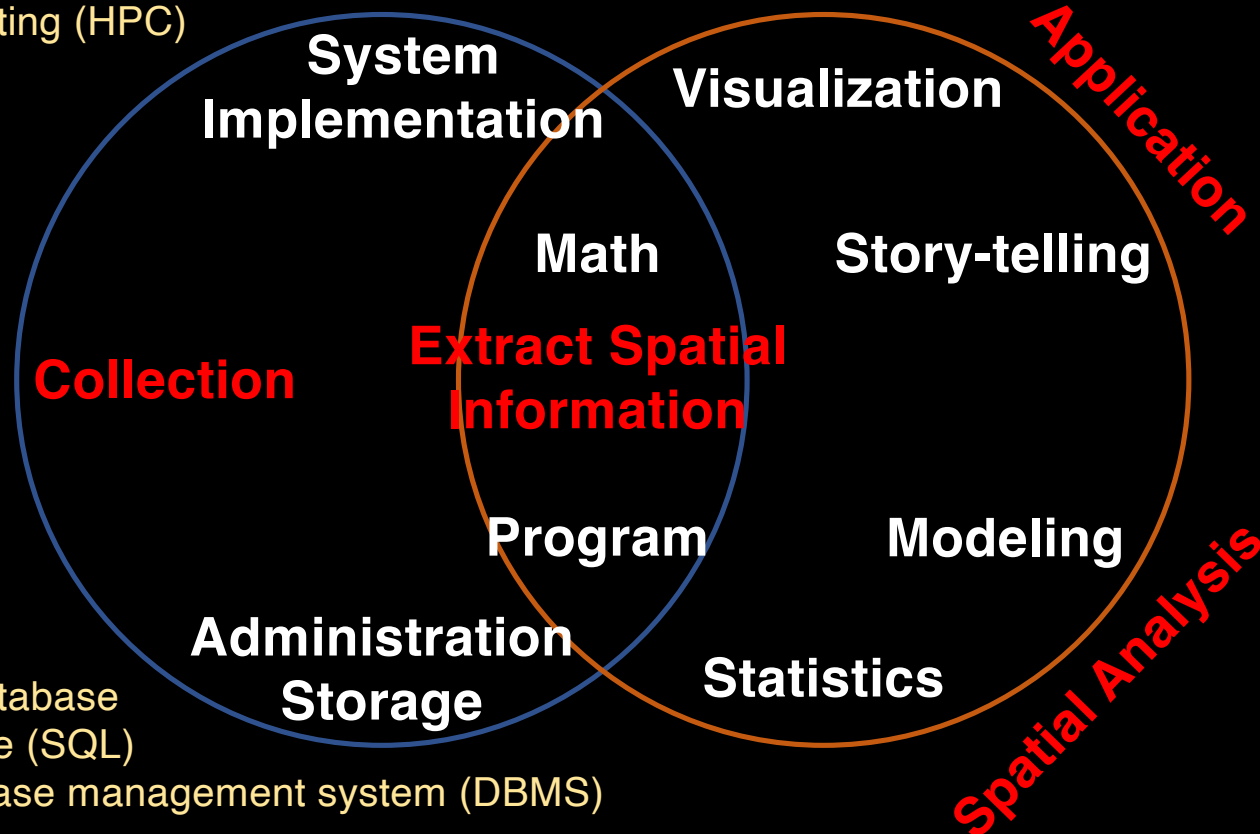
Geo-computing

High performance computing (HPC)
Cloud computing
Quantum computing
Computing algorithms

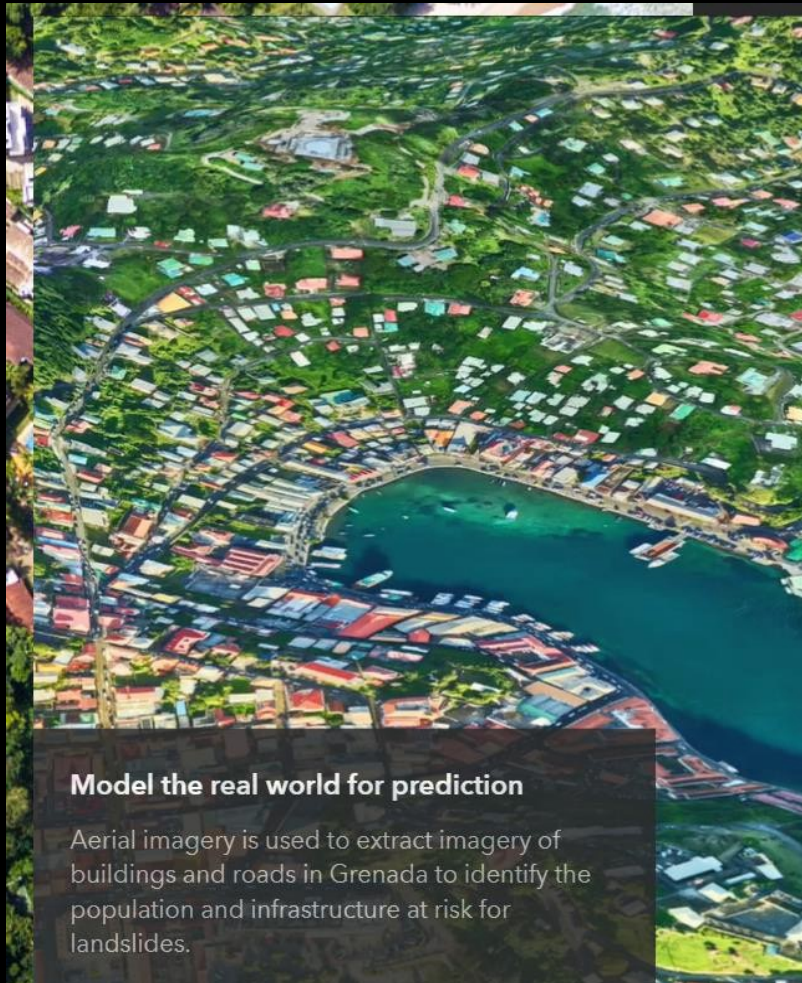
**Real
World**

Geo-database

Data structure
File and enterprise geodatabase
Structured query language (SQL)
Multiuser relational database management system (DBMS)



Geospatial Artificial Intelligence (GeoAI)



Model the real world for prediction

Aerial imagery is used to extract imagery of buildings and roads in Grenada to identify the population and infrastructure at risk for landslides.

DEFINITION

What is GeoAI?

Geospatial artificial intelligence (GeoAI) is the application of artificial intelligence (AI) fused with geospatial data, science, and technology to accelerate real-world understanding of business opportunities, environmental impacts, and operational risks. Organizations are modernizing operations to run at scale through automated data generation and approachable spatial tools and algorithms.

01. Extract rich geospatial data with deep learning

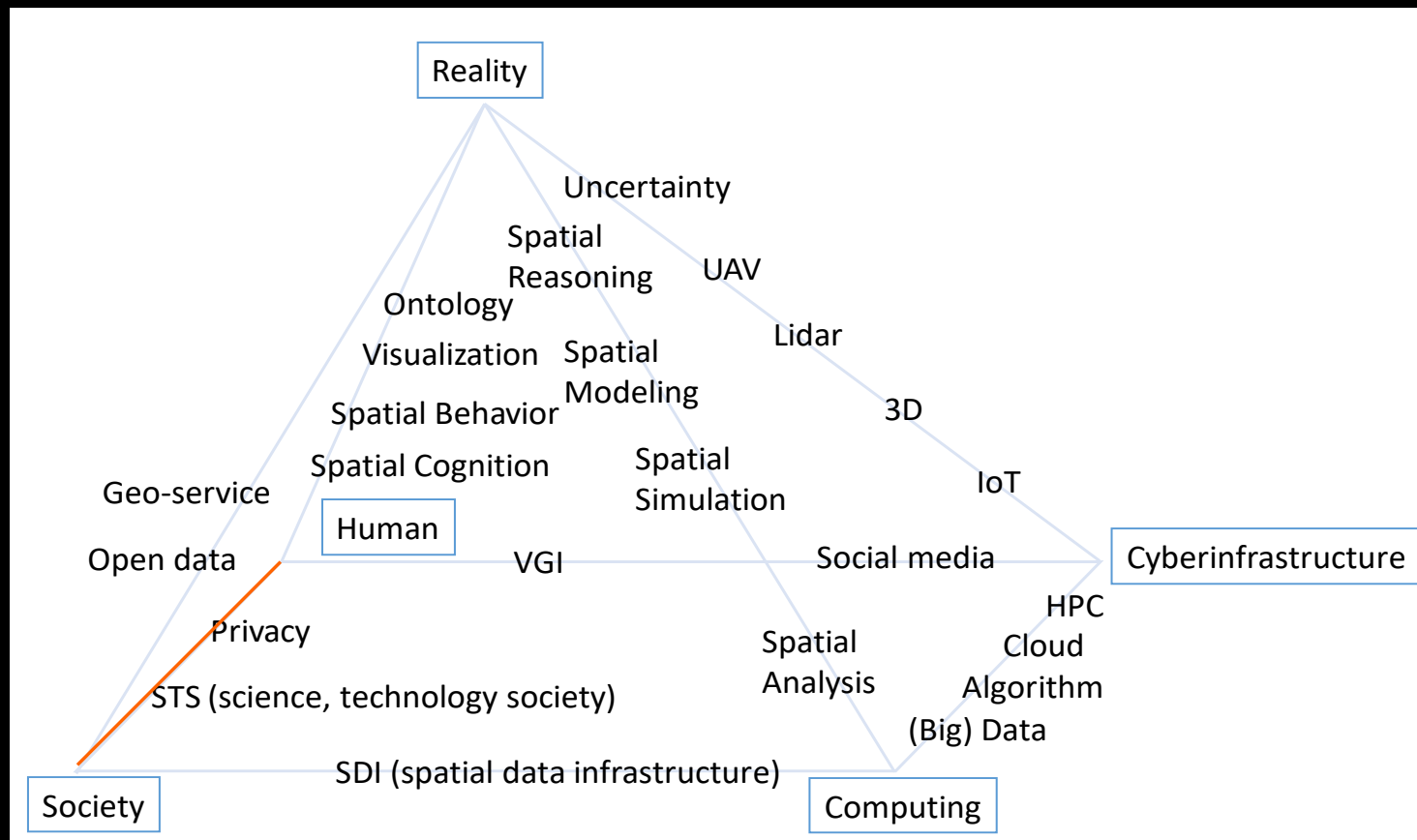
Save time by automating the extraction, classification, and detection of information from data such as imagery, video, point clouds, and text.

02. learning

Build more accurate models. Detect clusters, calculate change, find patterns, and forecast outcomes with spatial algorithms backed by experts.

New Era of GIS

- GIS = GI Science = Geographic Information Science

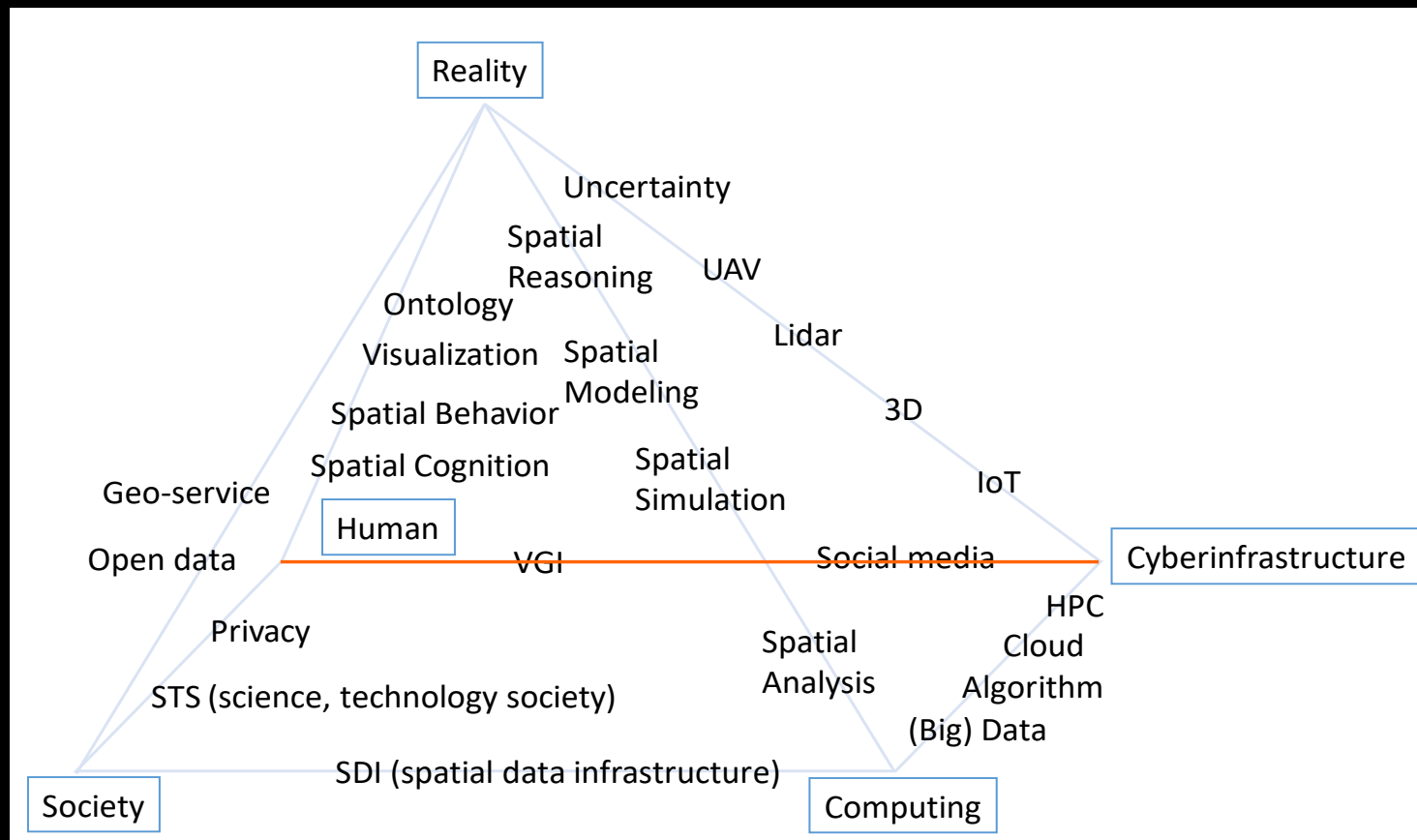


Ethics, power, and critical GIS

- Who creates data?
- Who is represented or excluded?
- Has geo-privacy been addressed?

New Era of GIS

- GIS = GI Science = Geographic Information Science



Spatial Big Data

- Volunteered Geographic Information (VGI)
- Social media
- Real-time data (cellphone, transportation data, etc.)

18. 某生利用高速公路電子收費系統在通勤時段的流量資料，探究臺灣地區民眾的活動範圍。他將汽車進入與離開高速公路的起迄資料繪製於地圖上，並根據資料在空間上的密集程度，定義數個生活圈。圖 3 是起迄資料與生活圈的呈現結果。下列何者最適合成為該探究活動的標題？

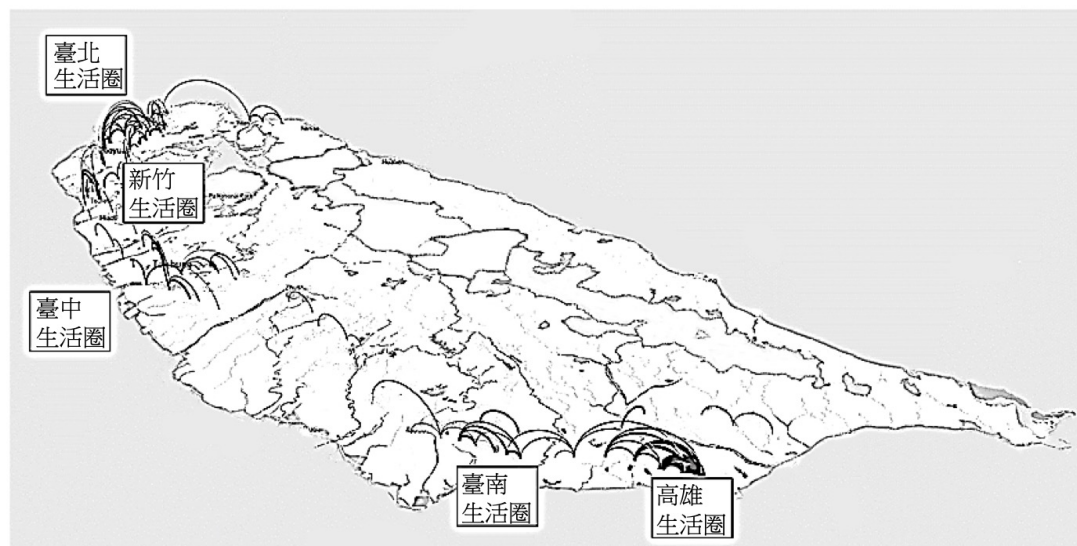
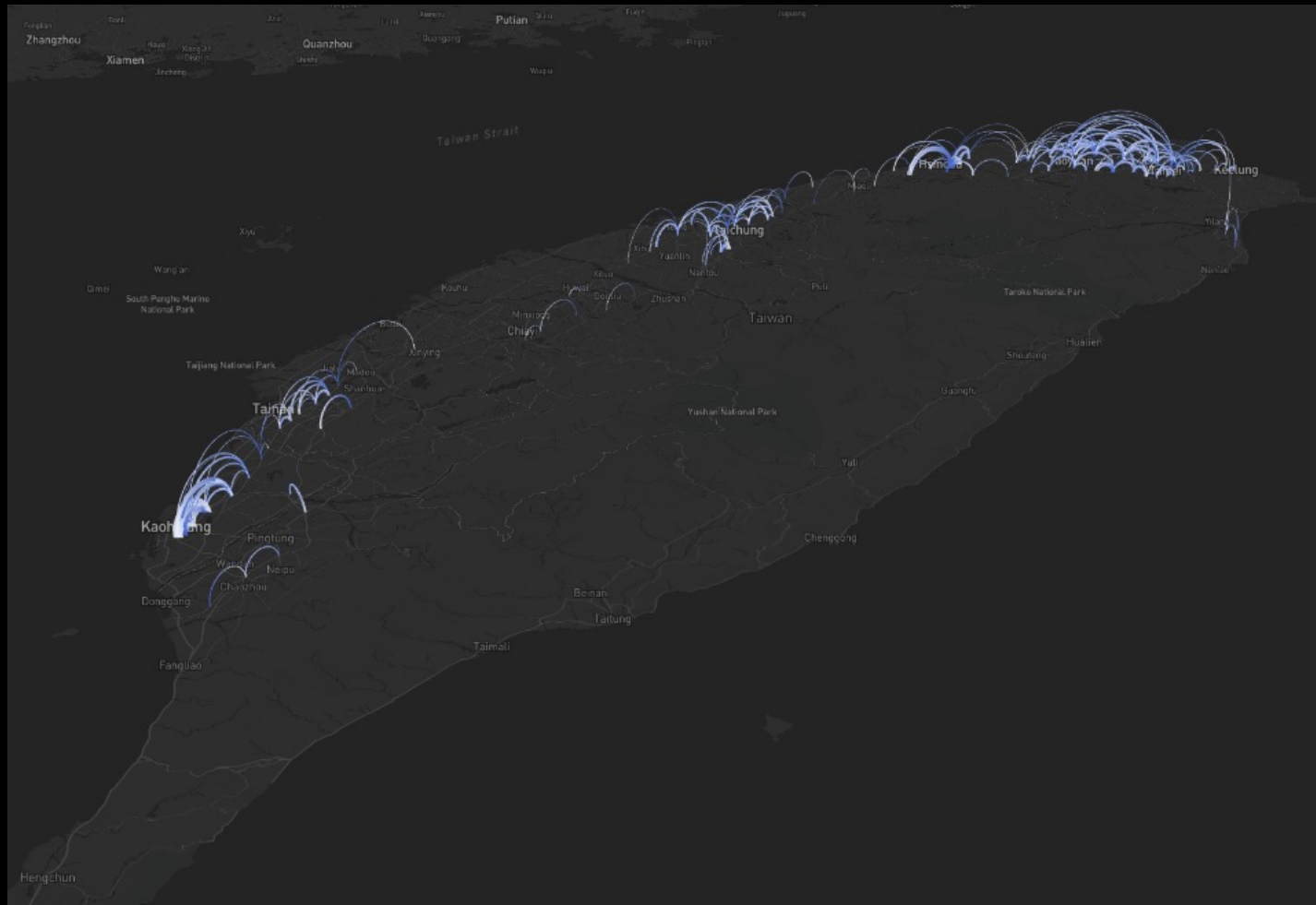


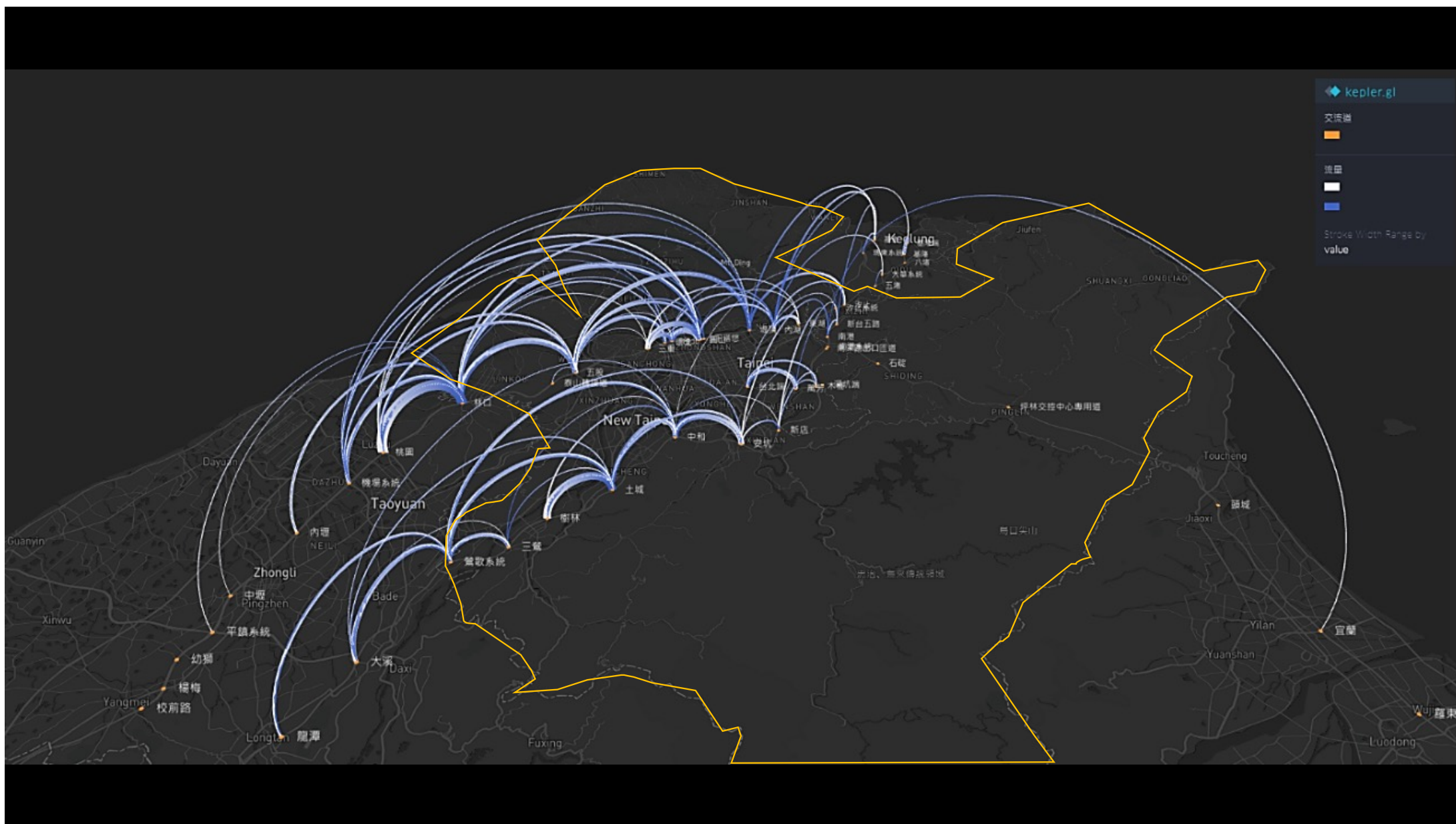
圖 3

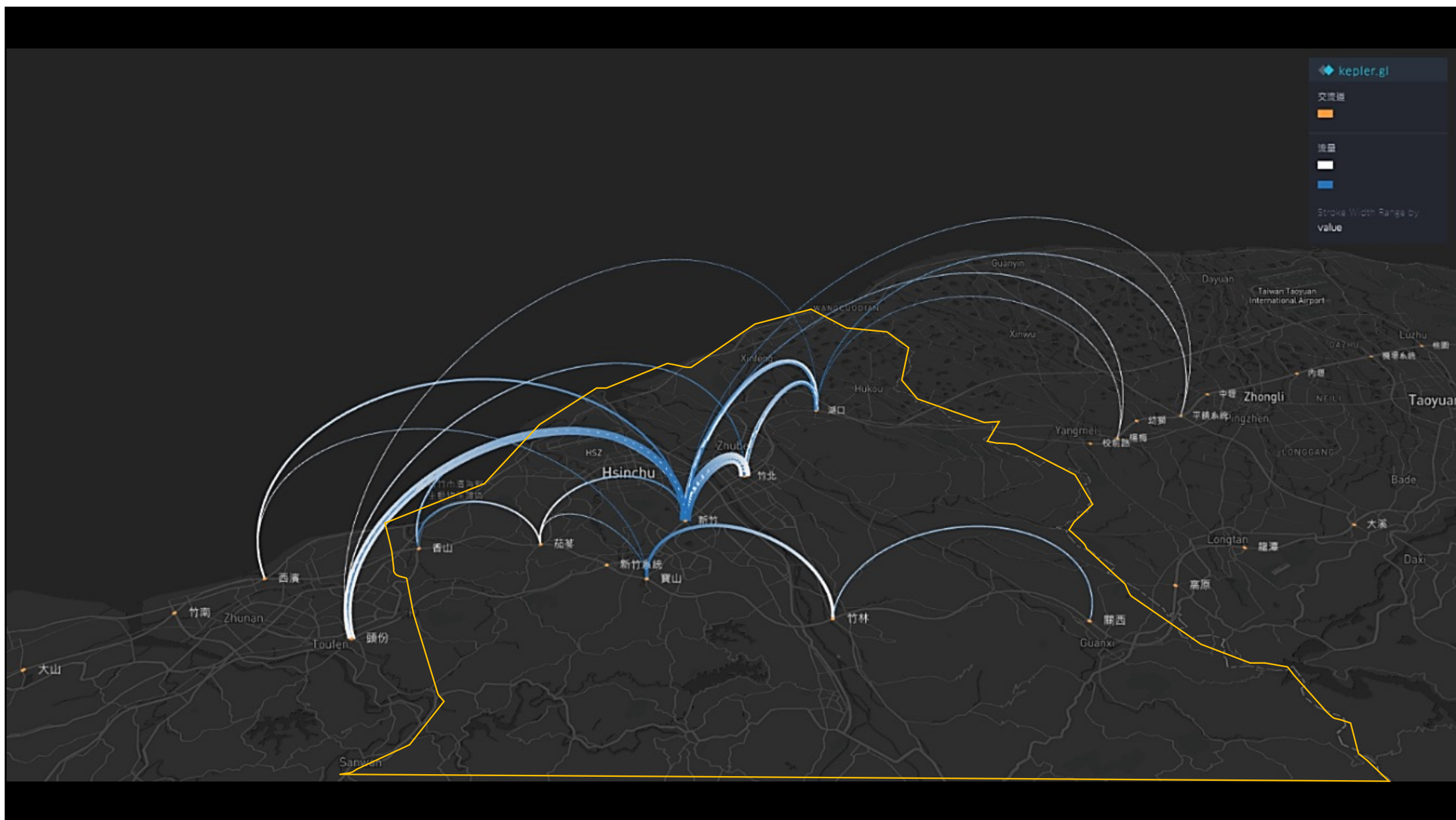
- | | |
|------------------|------------------|
| (A) 資訊革命如何改變生活模式 | (B) 運輸活動如何影響聚落系統 |
| (C) 路網型態如何加速人口遷移 | (D) 交通革新如何造就都市機能 |

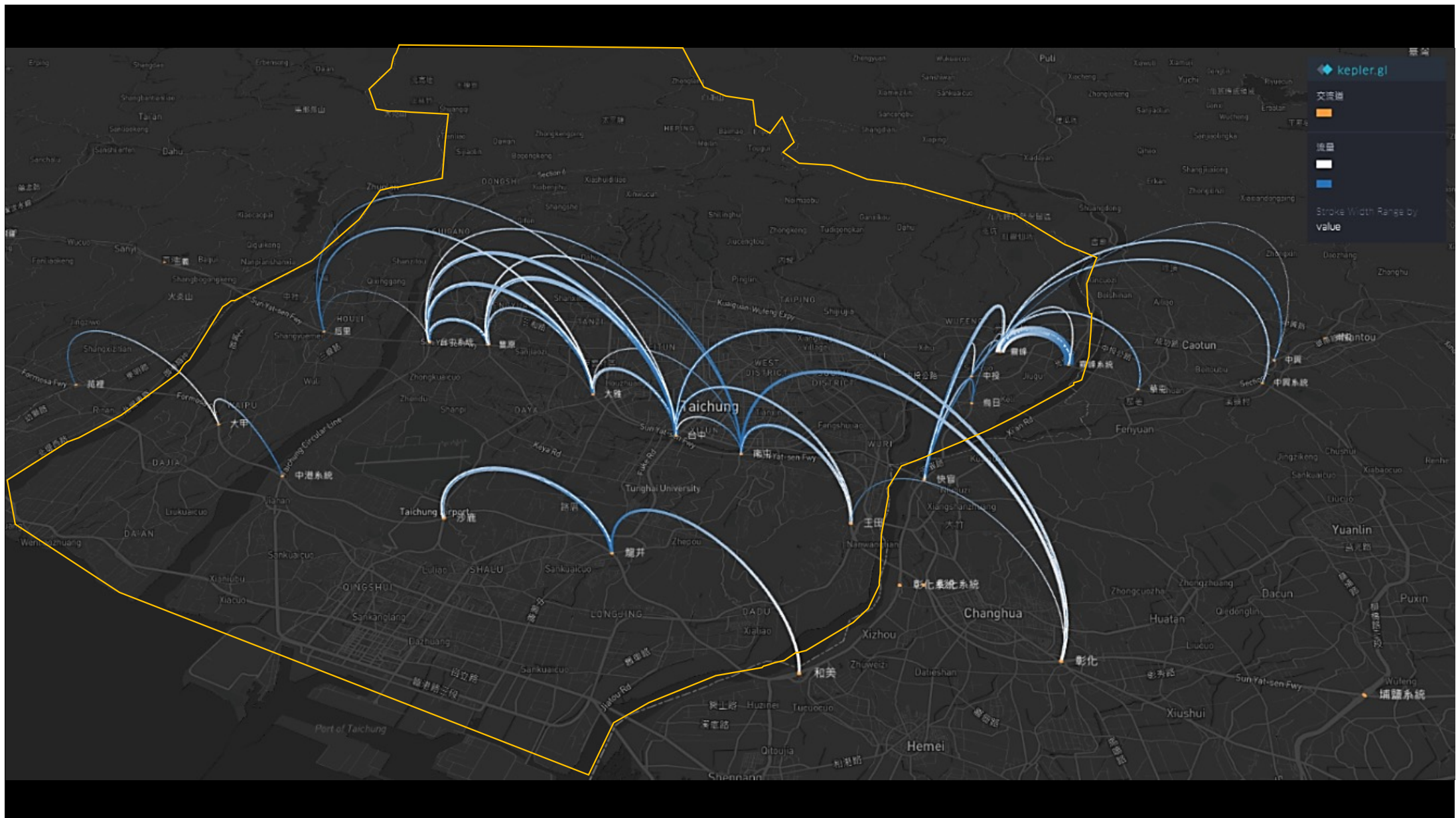
Job-Housing Balance

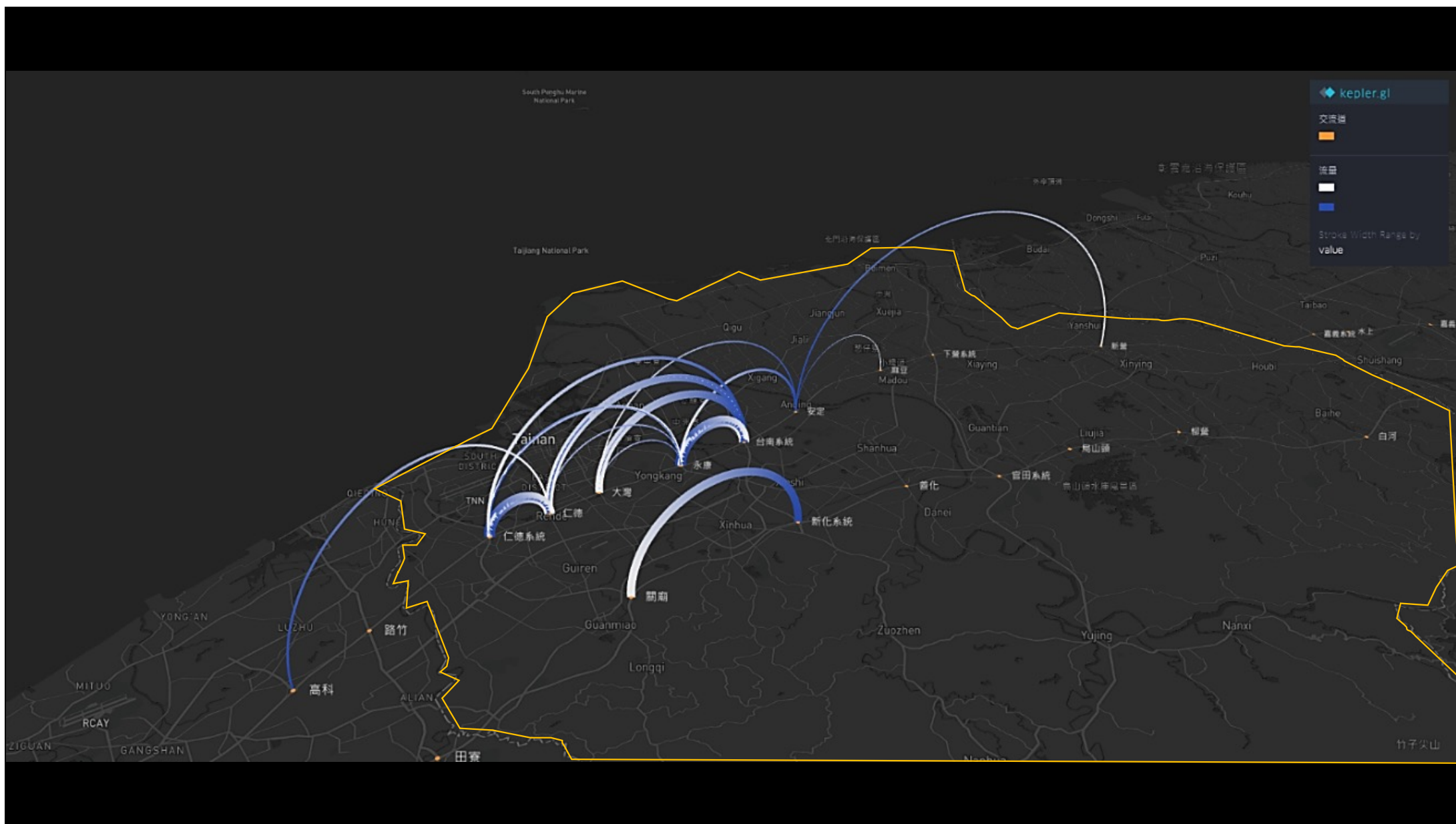


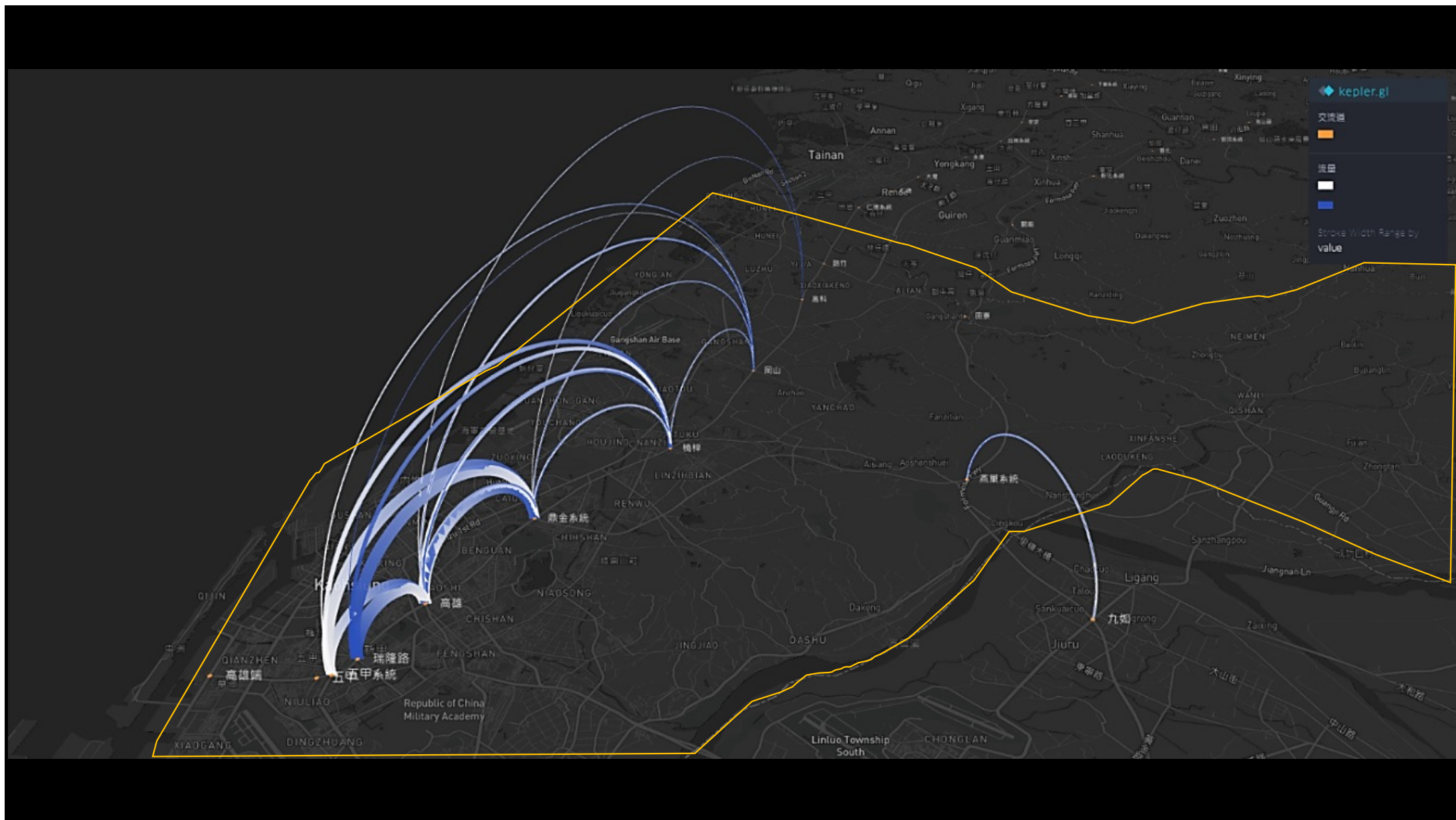
介姿淇 (2022)



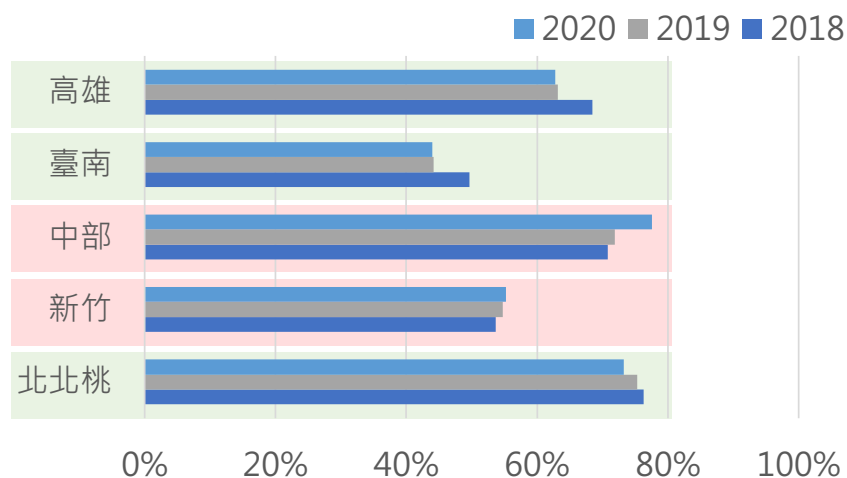








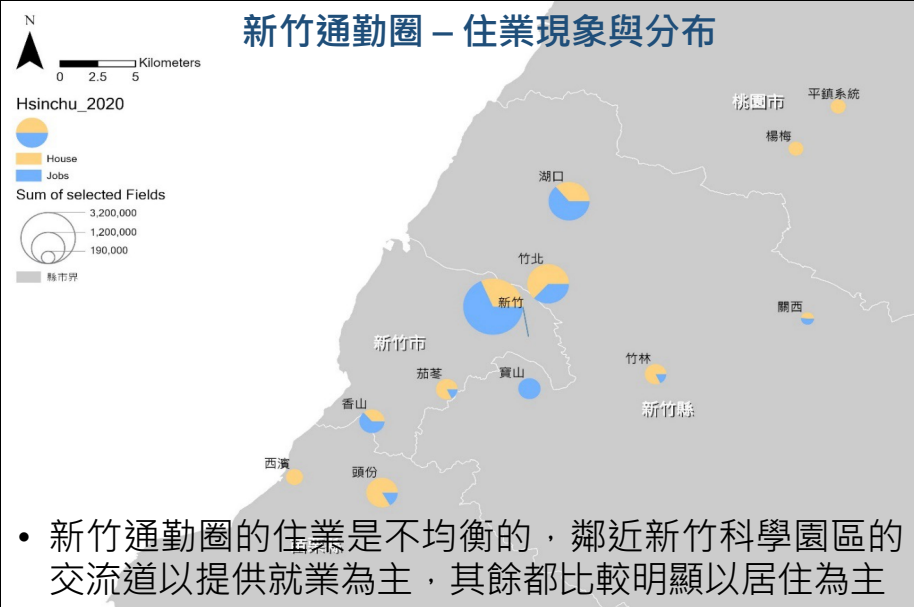
Excess Commuting (超額通勤)



- 北北桃通勤圈、高雄通勤圈、臺南通勤圈逐年有些微降低的趨勢
- 新竹通勤圈以及中部通勤圈則有逐年增加的趨勢

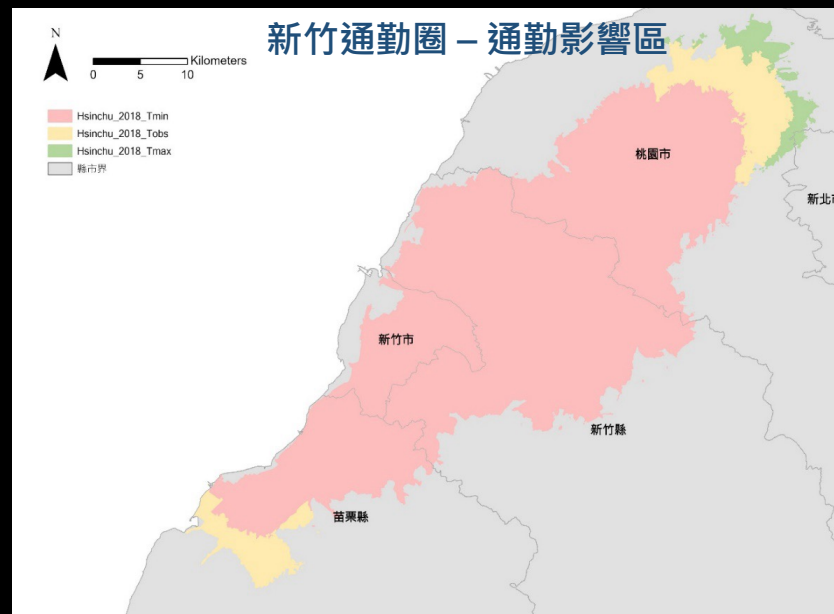
- 新竹通勤圈擴及桃園與苗栗頭份、竹南，將來可能繼續往北擴散
- 北北桃與新竹通勤圈範圍有重疊的，將來有合併的趨勢

新竹通勤圈 – 住業現象與分布

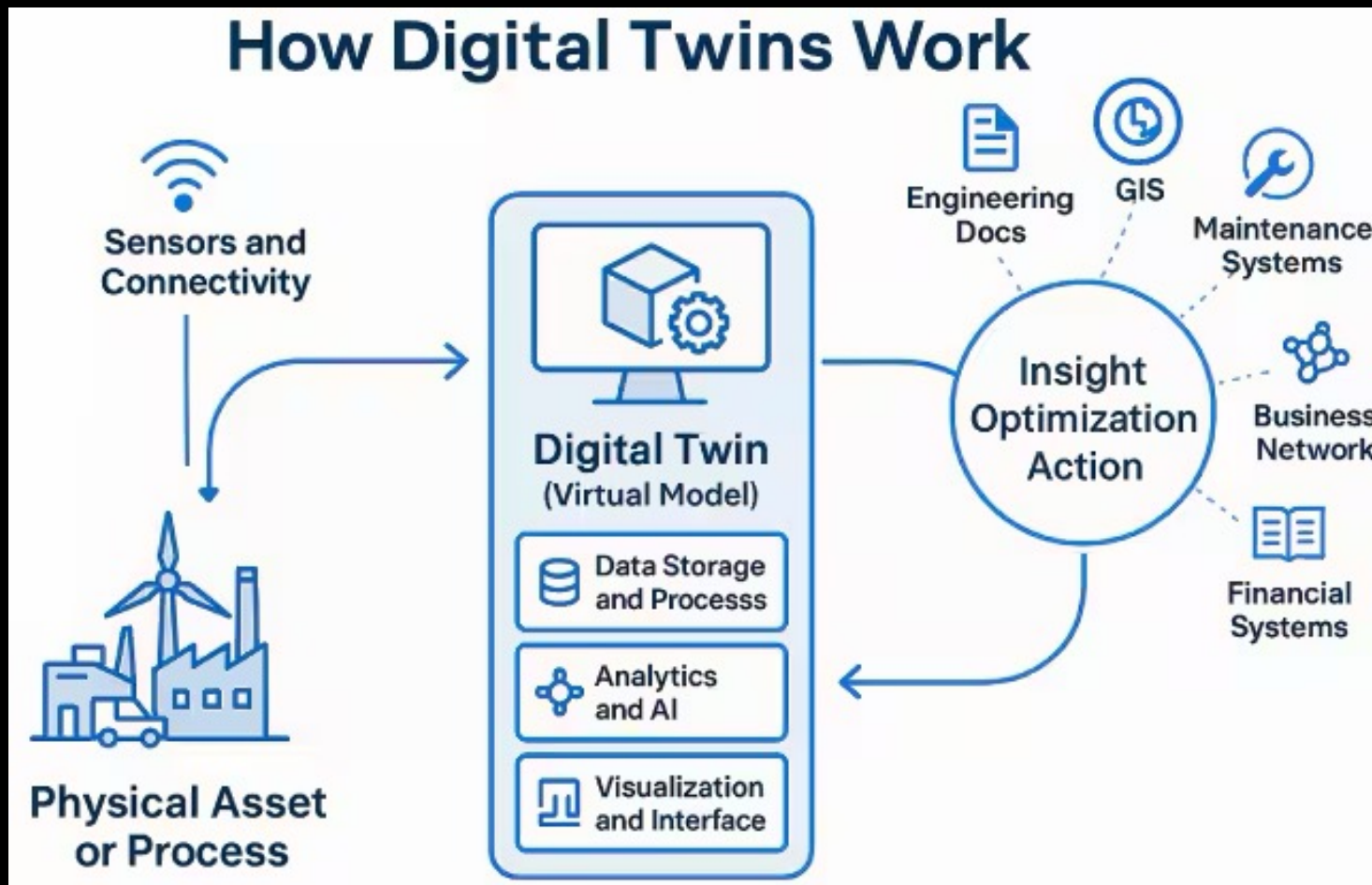


- 新竹通勤圈的住業是不均衡的，鄰近新竹科學園區的交流道以提供就業為主，其餘都比較明顯以居住為主

新竹通勤圈 – 通勤影響區



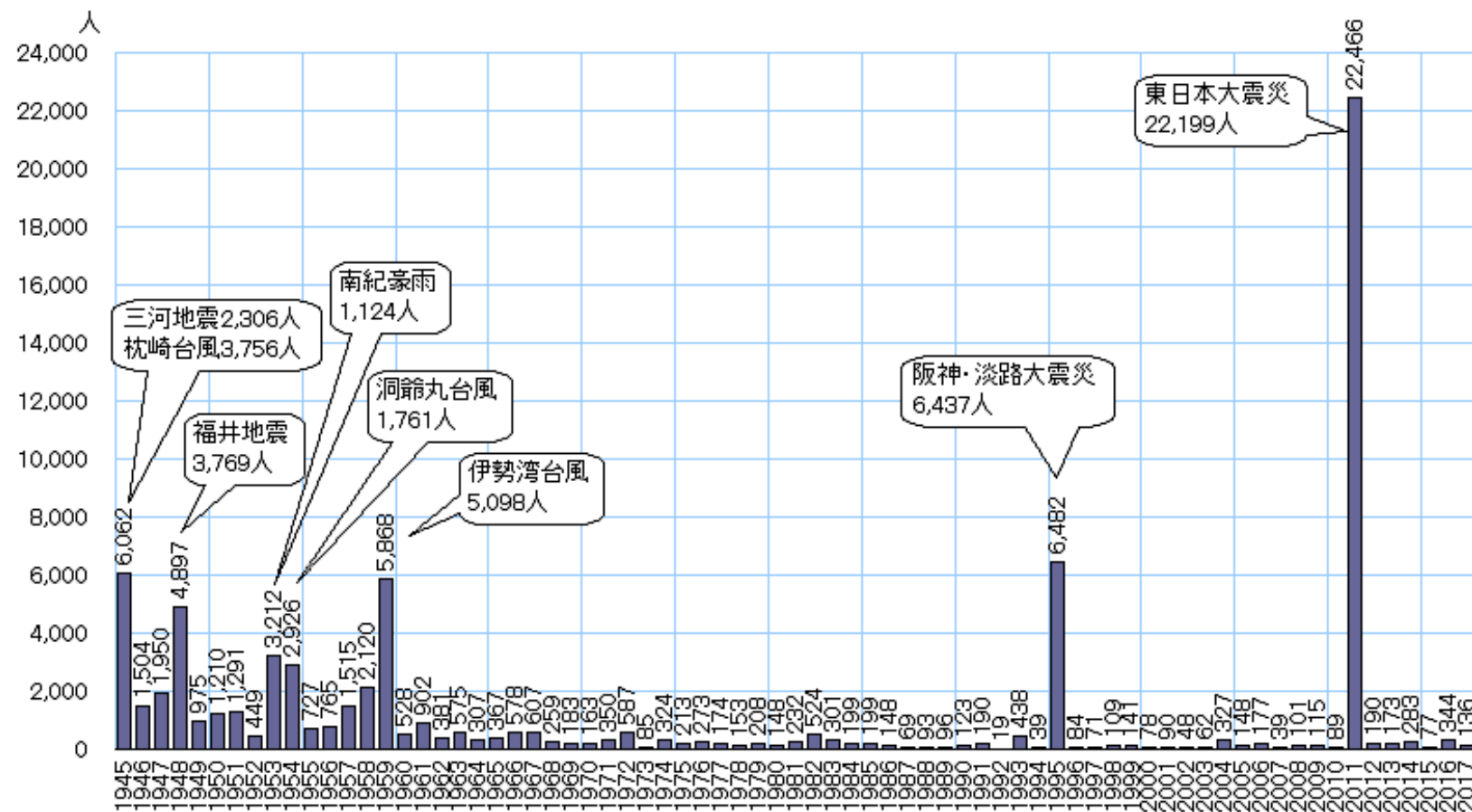
Digital Twin (數位孿生)



Problem-Solving GIScience

- GIScience is to solve spatial problems
- Pay attention to “geographic information” when you use GIS
- Require multiple approaches and cross-disciplinary knowledge to solve current geographic problems

自然災害による死者・行方不明者数の推移

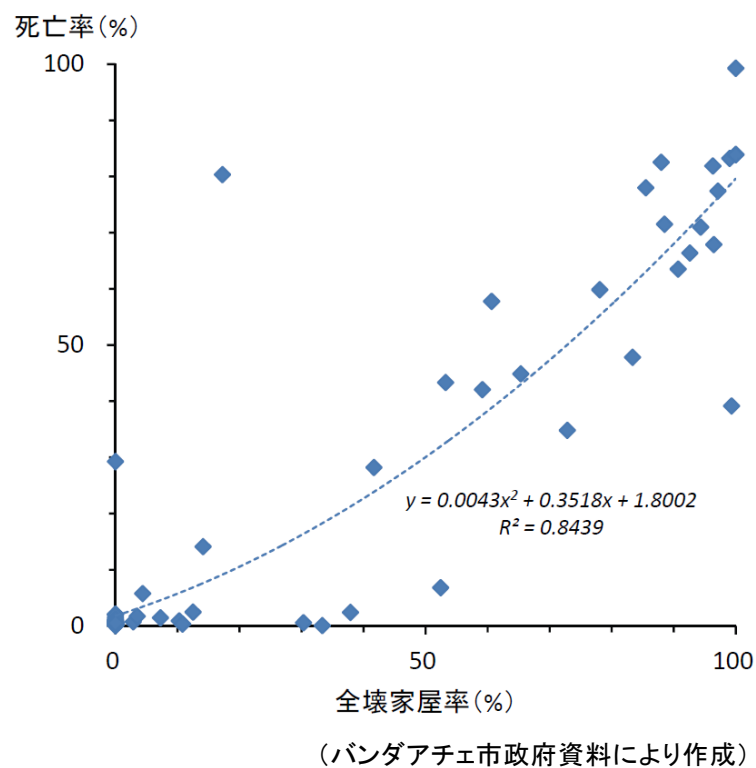


(注) 1945年は主な災害による死者・行方不明者(理科年表による)。46～52年は日本気象災害年報, 53～62年は警察庁資料, 63年以降は消防庁資料に基づき内閣府作成。1995年の死者のうち、阪神・淡路大震災の死者については、いわゆる関連死919名を含む(兵庫県資料)。2017年は内閣府とりまとめによる速報

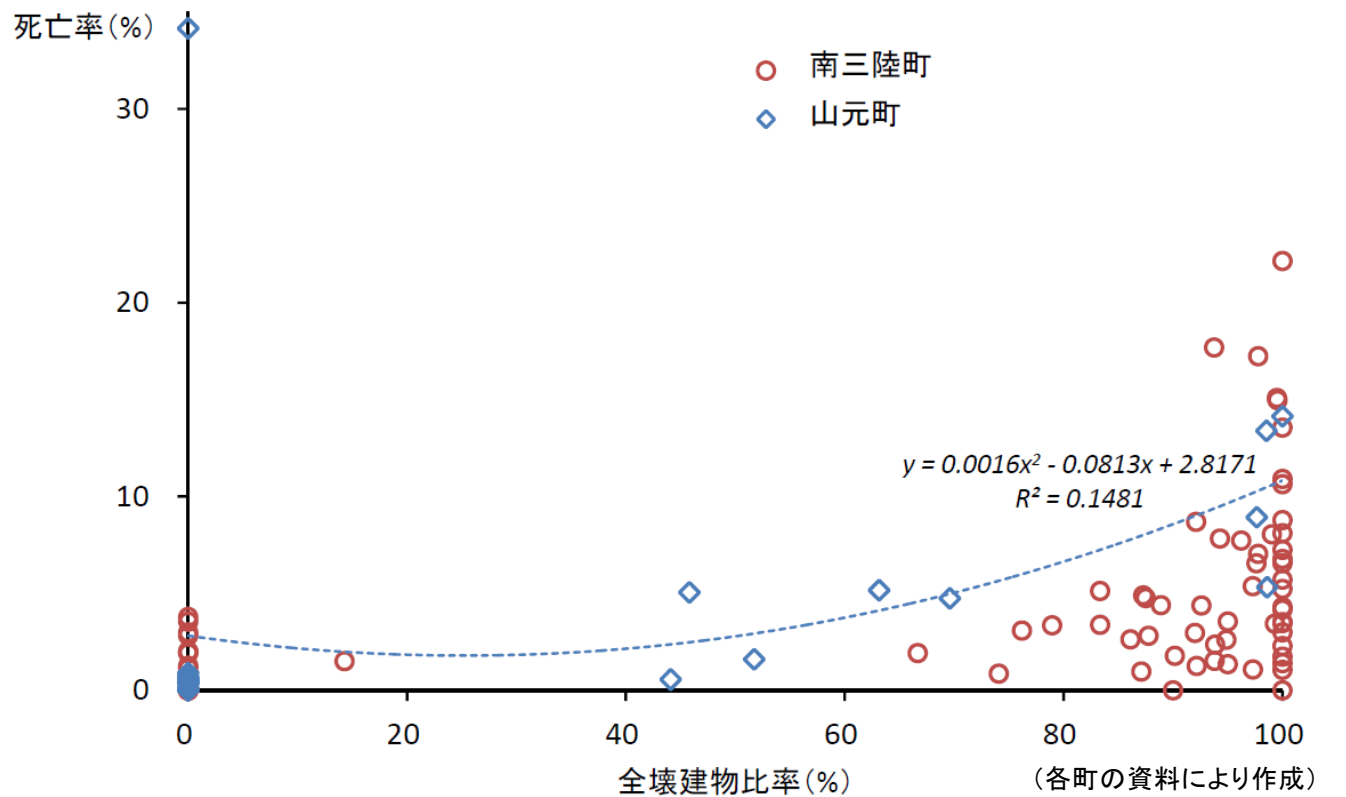
(資料) 内閣府「平成30年版防災白書」ほか



2004年南亞大海嘯印尼班達亞齊的受災情況

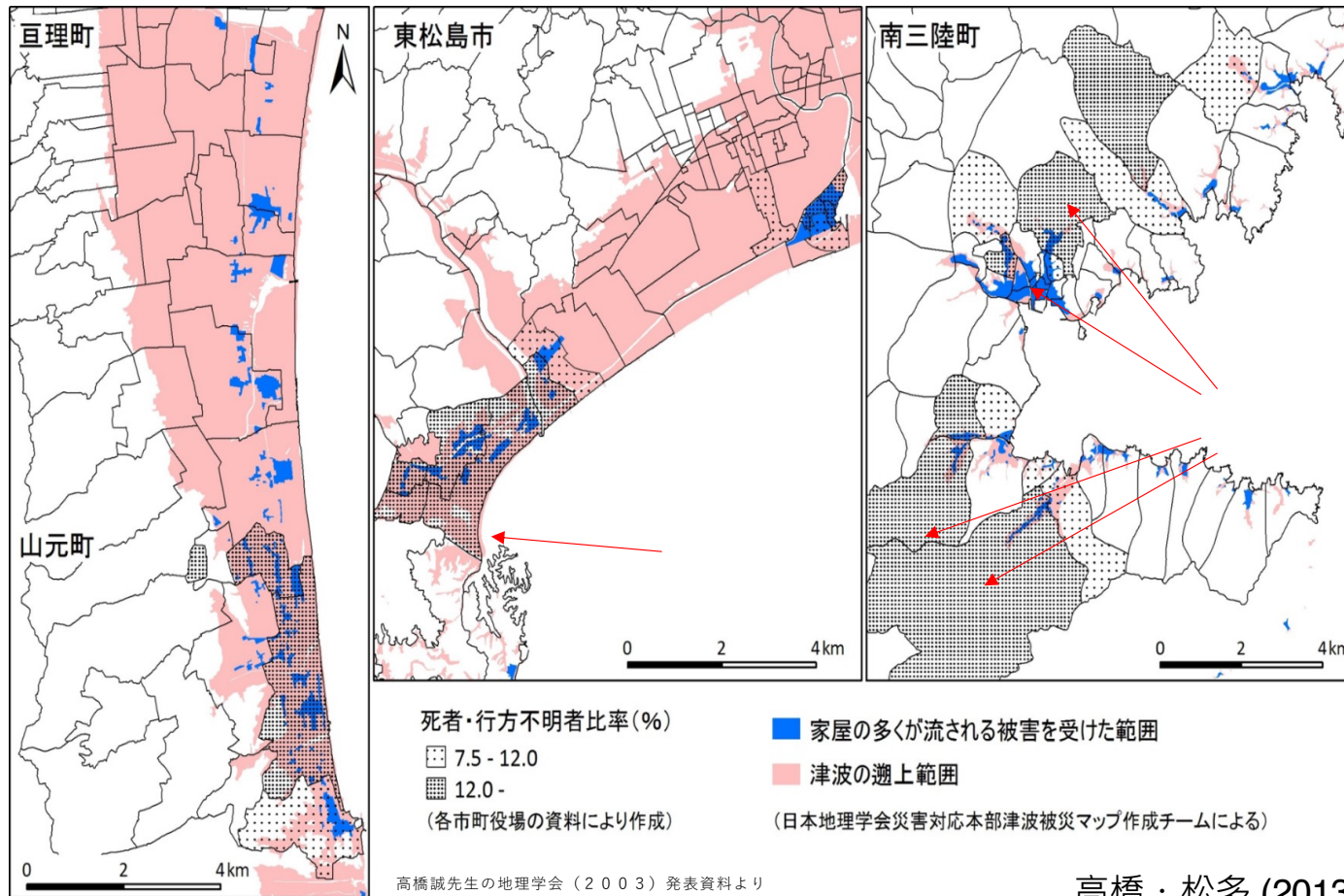


2011年東日本大震災海嘯南三陸町與山元町の受災情況



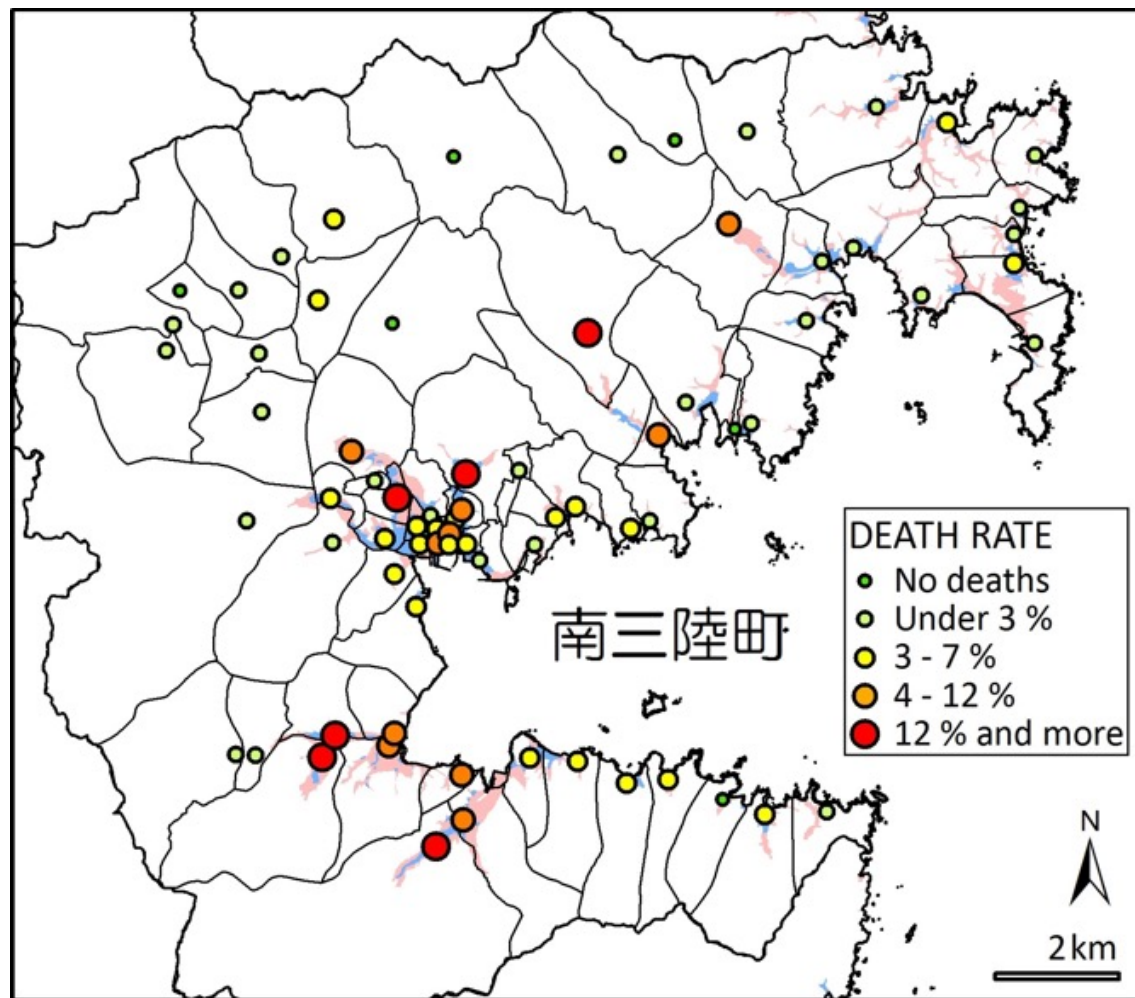
高橋・松多 (2013)

高死亡率の地域



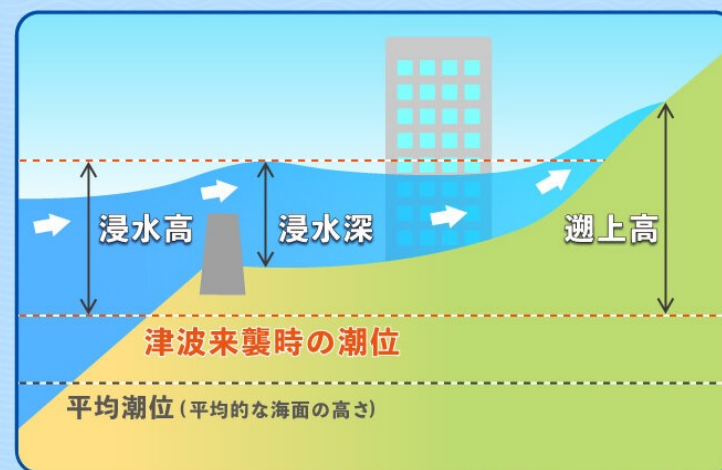
高橋誠先生の地理学会（2003）発表資料より

高橋・松多 (2013)



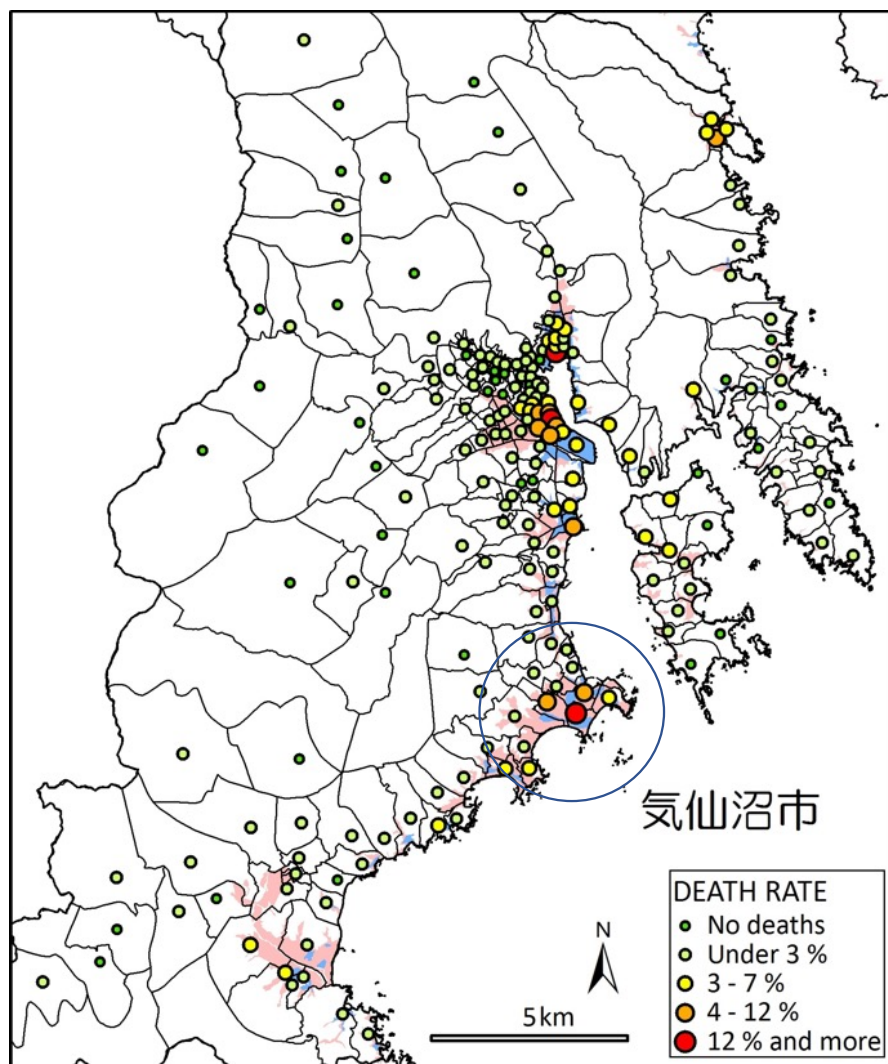
津波の影響する範囲

tenki.jp
知る防災

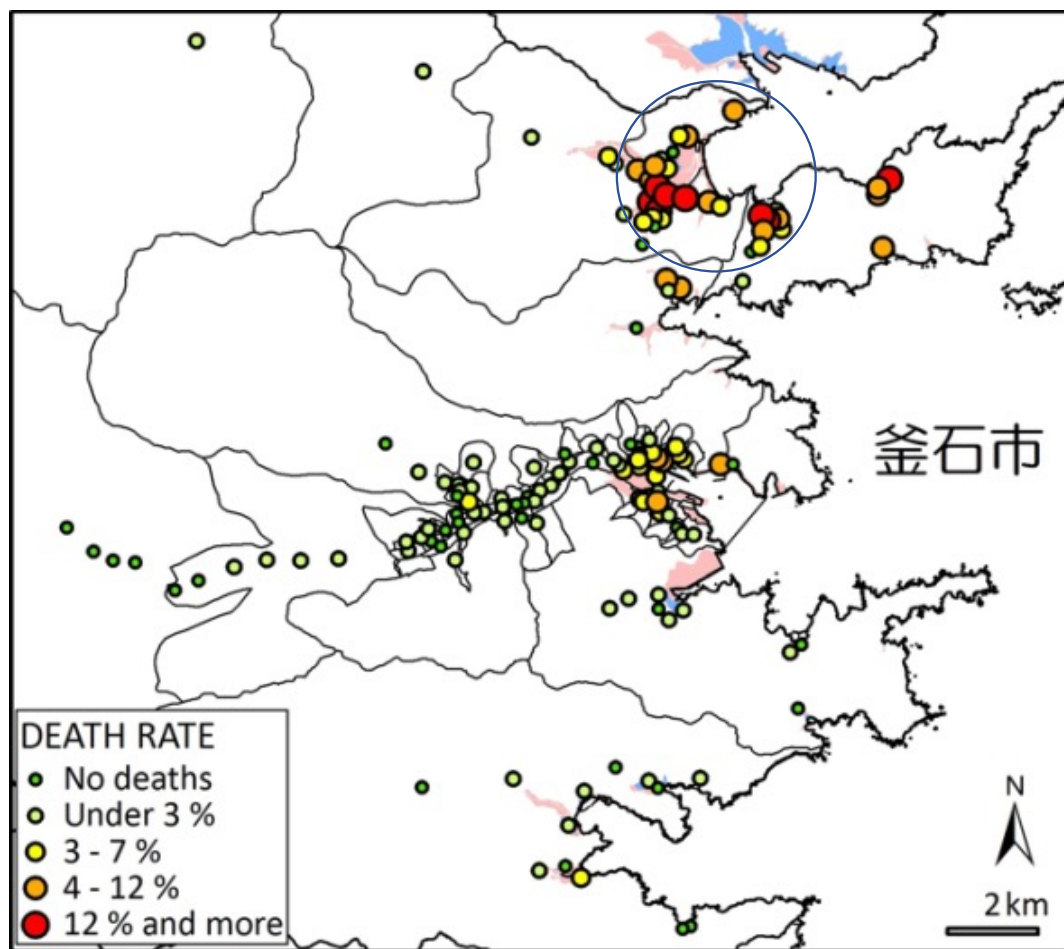


南三陸町戸倉小學校

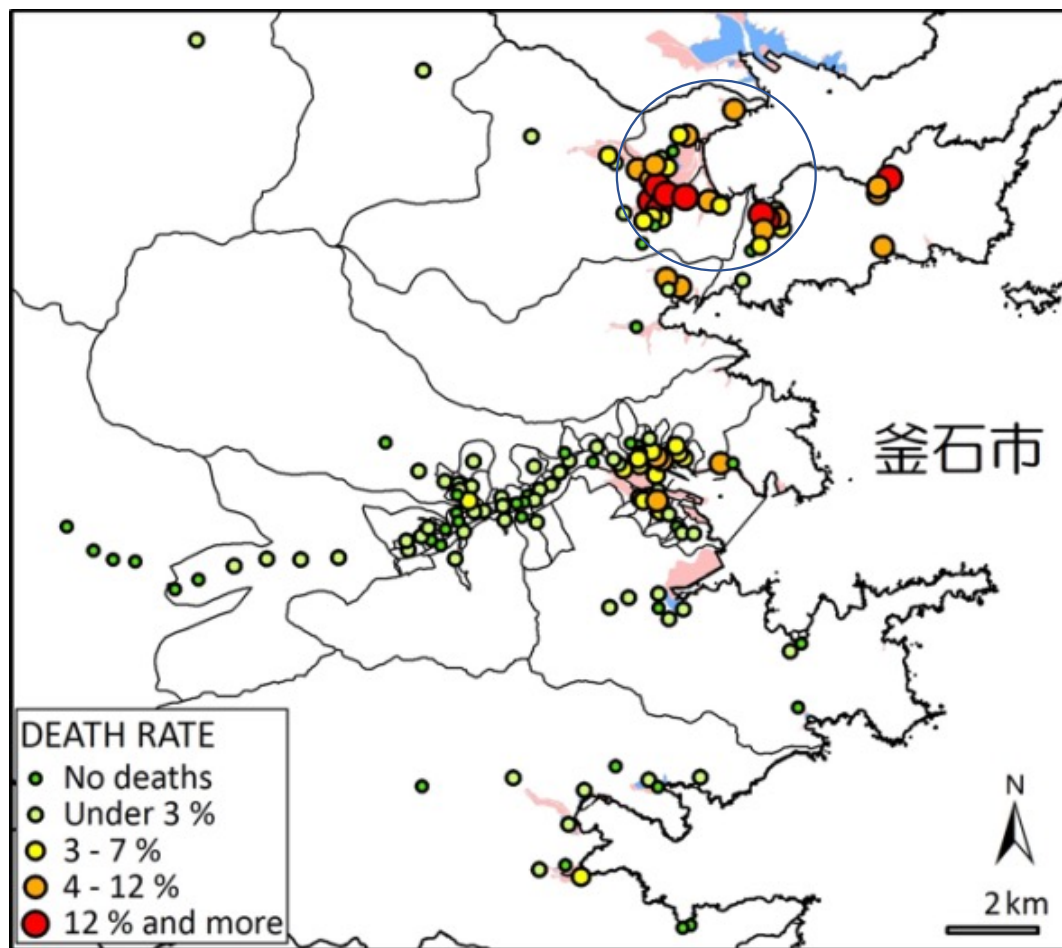




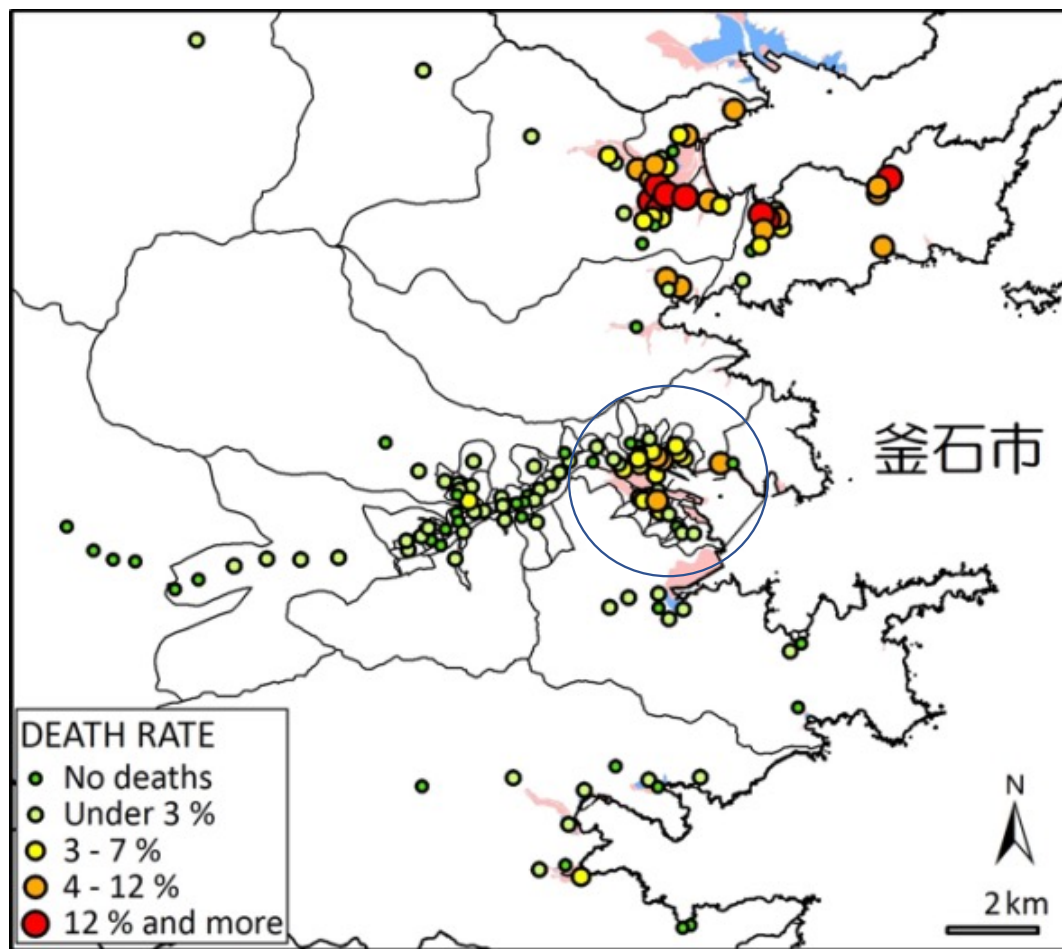
釜石市鵜住居地区



釜石市鵜住居地區



釜石市大川地區



石巻市立大川小學校





